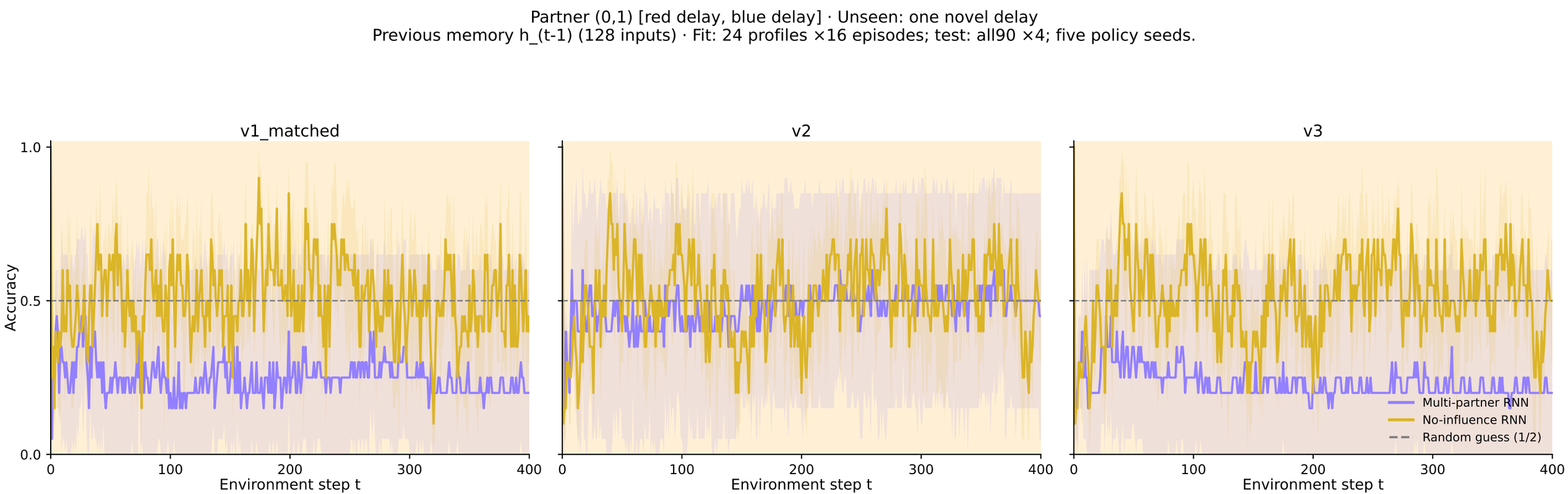
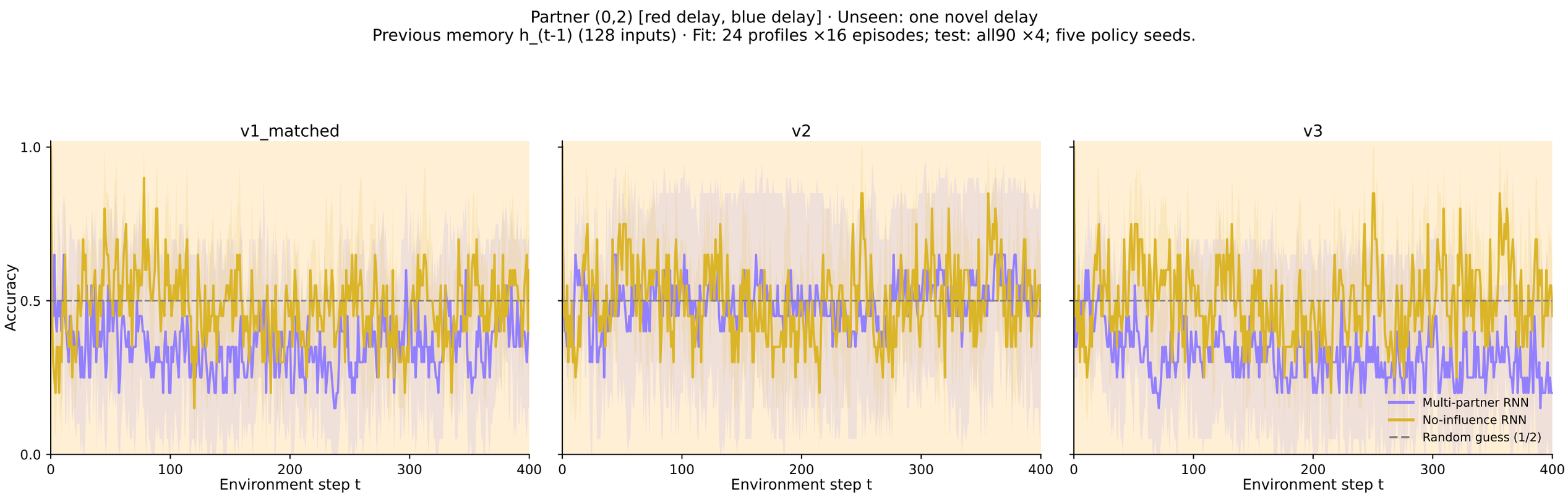
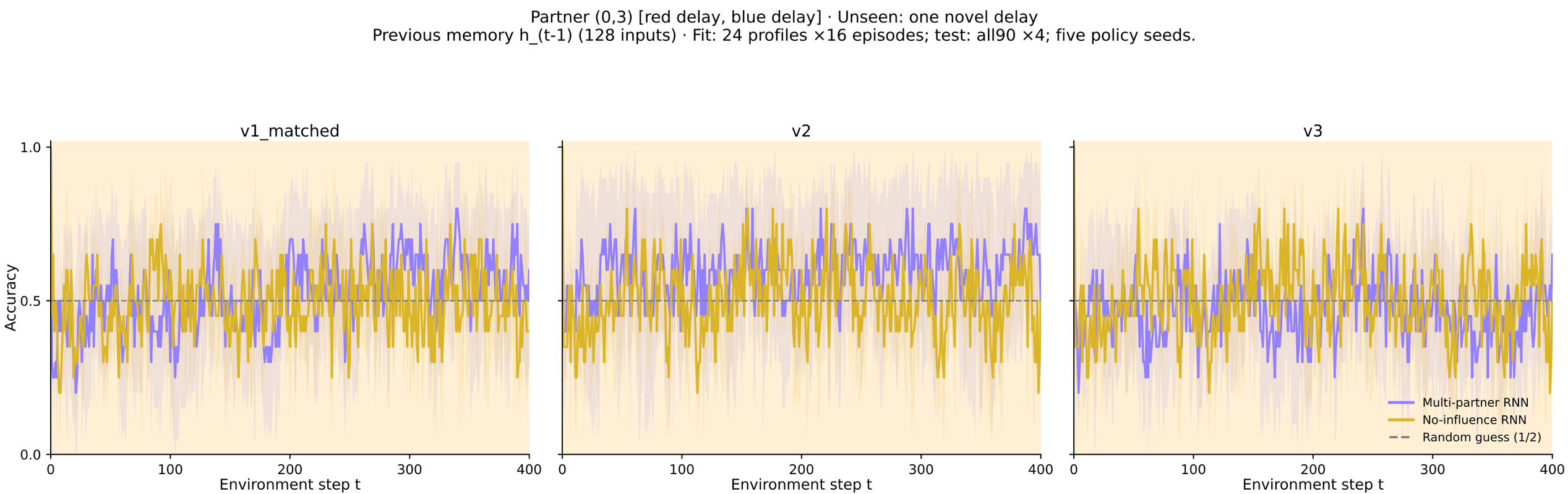
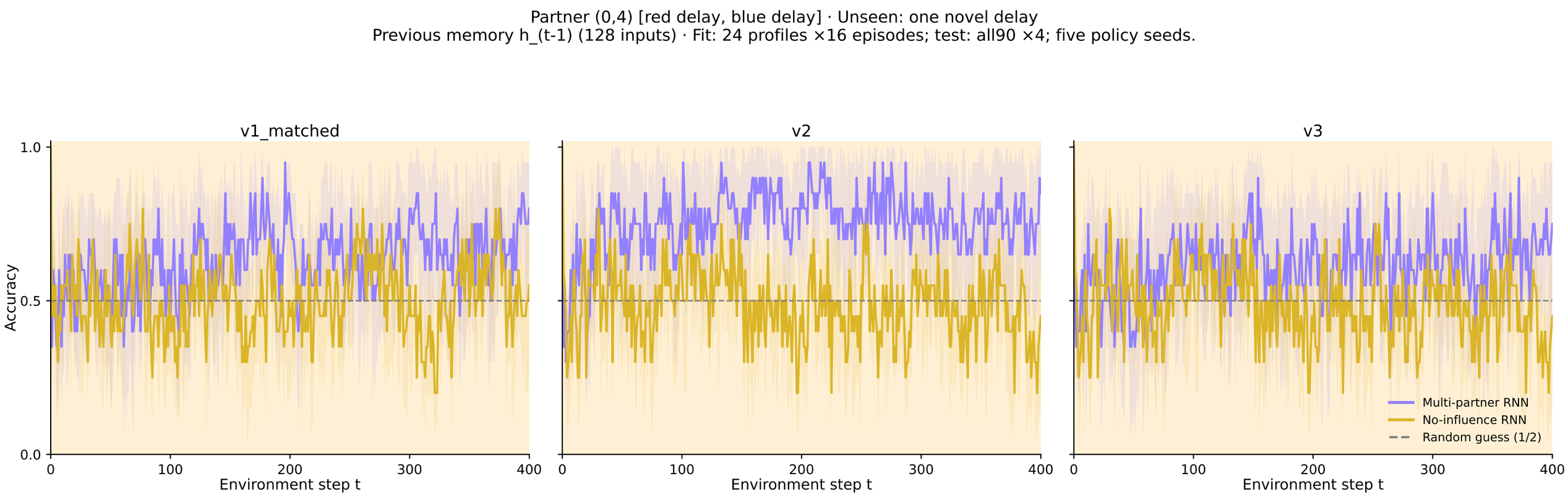
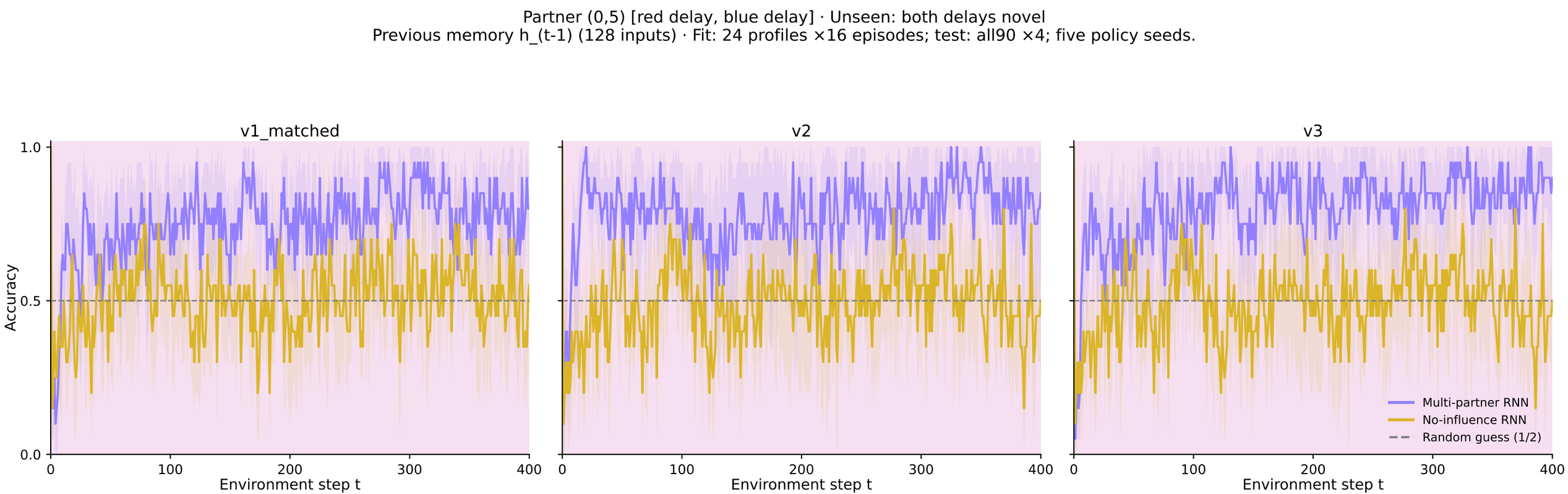
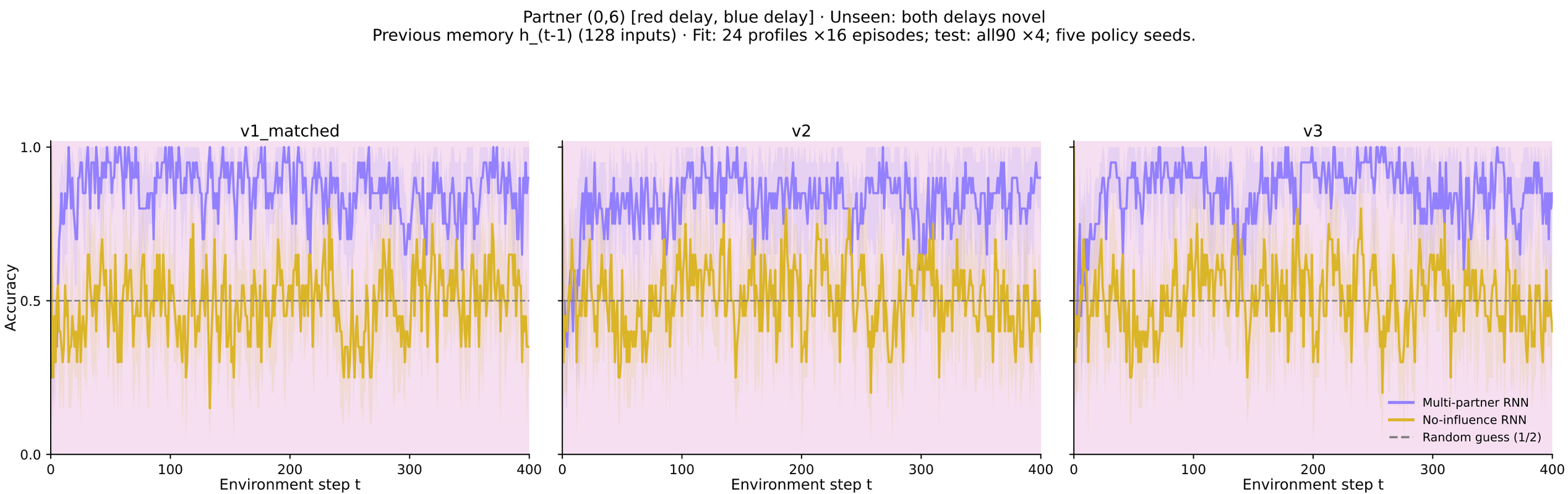

Partner (0,1) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

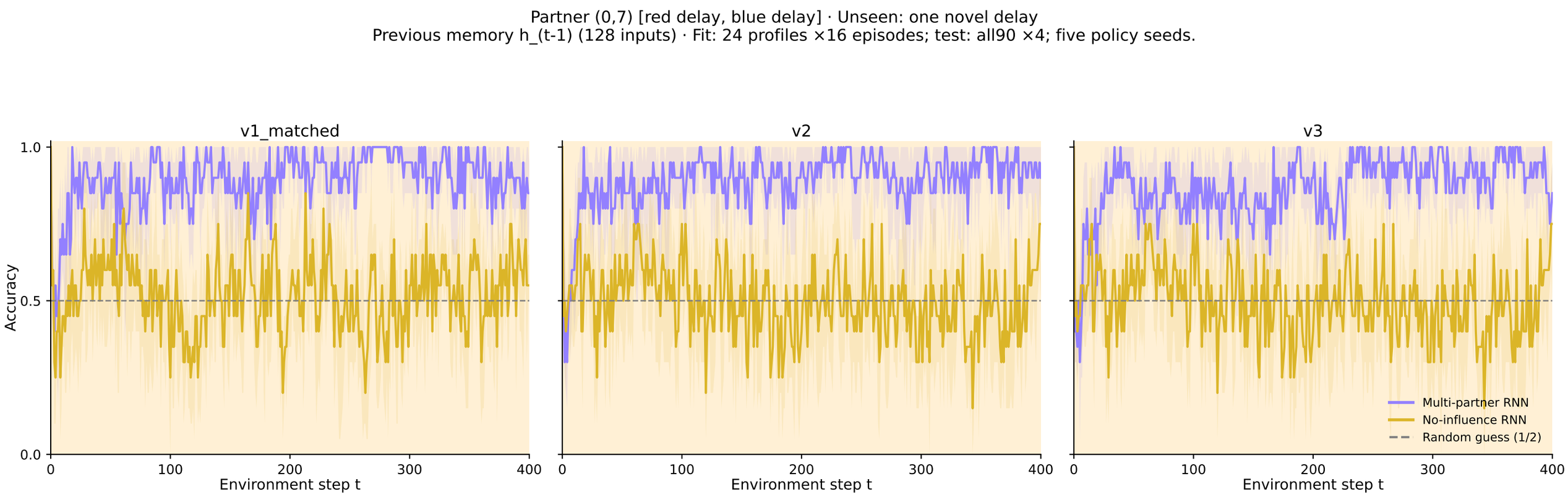

Partner (0,2) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

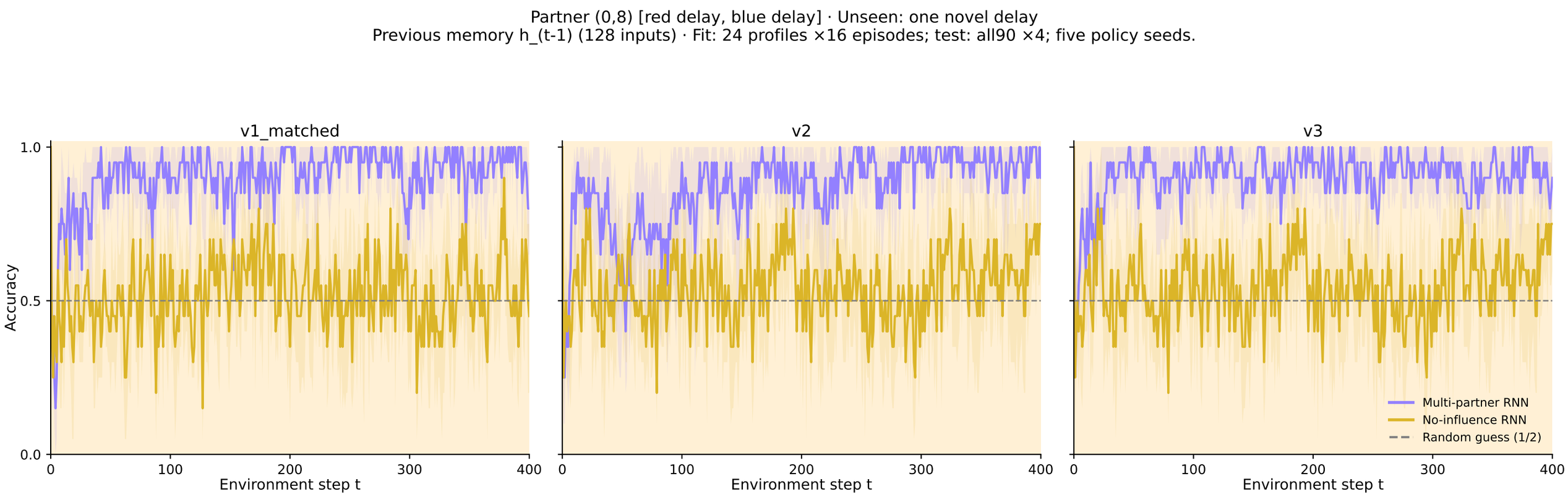

Partner (0,3) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

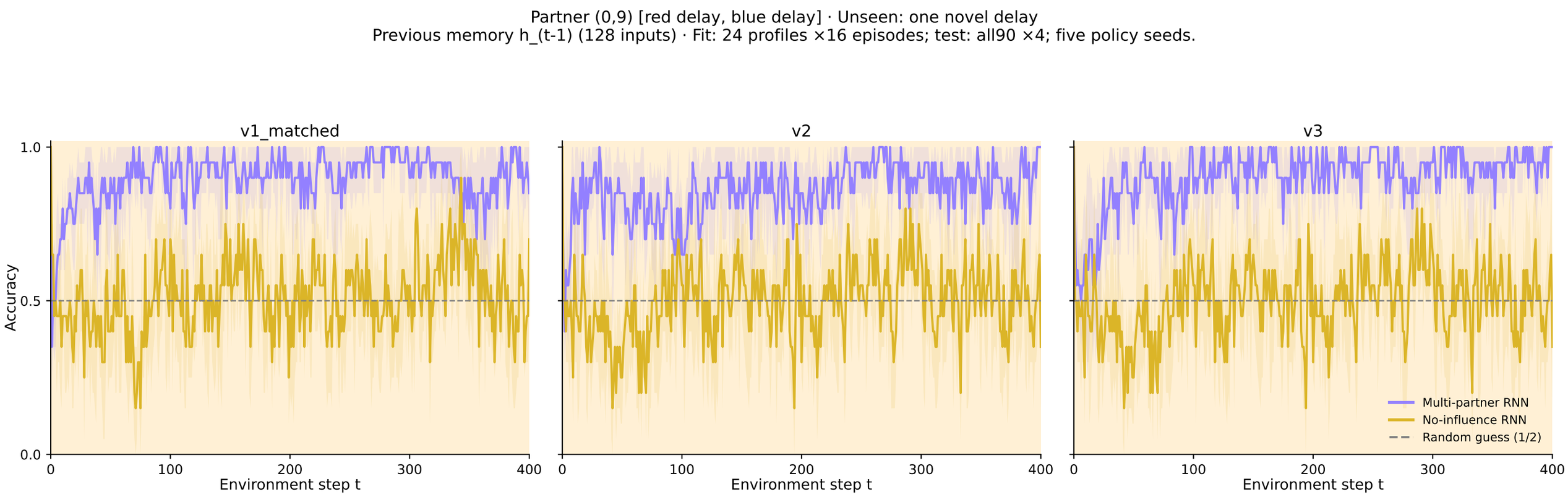

Partner (0,4) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

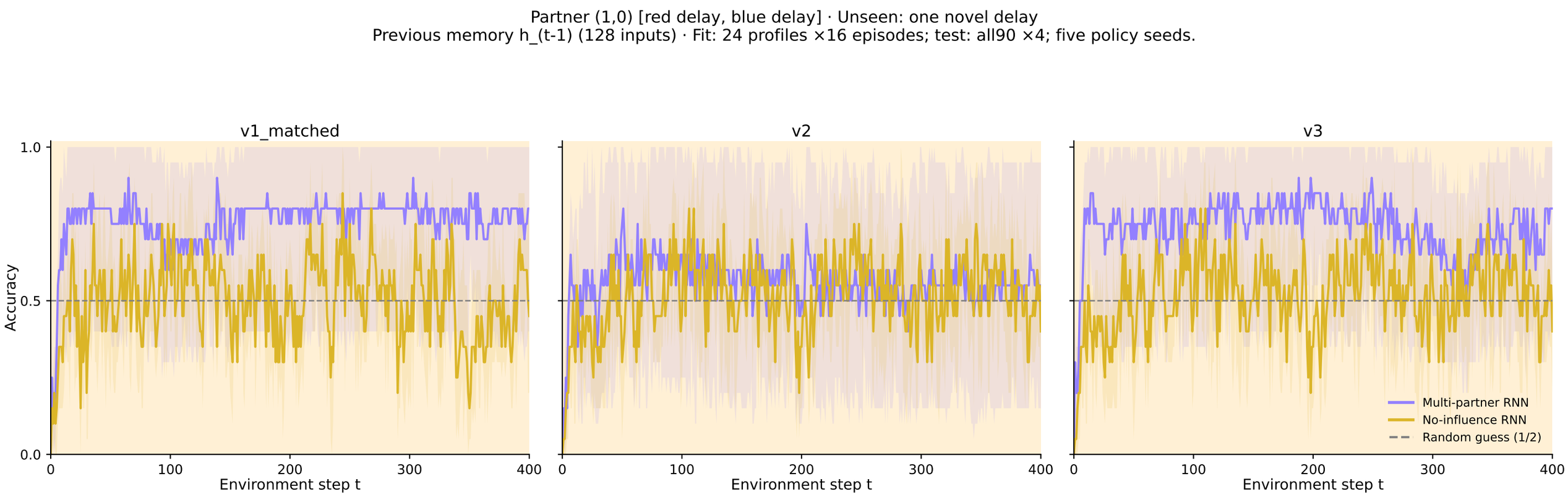

Partner (0,5) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

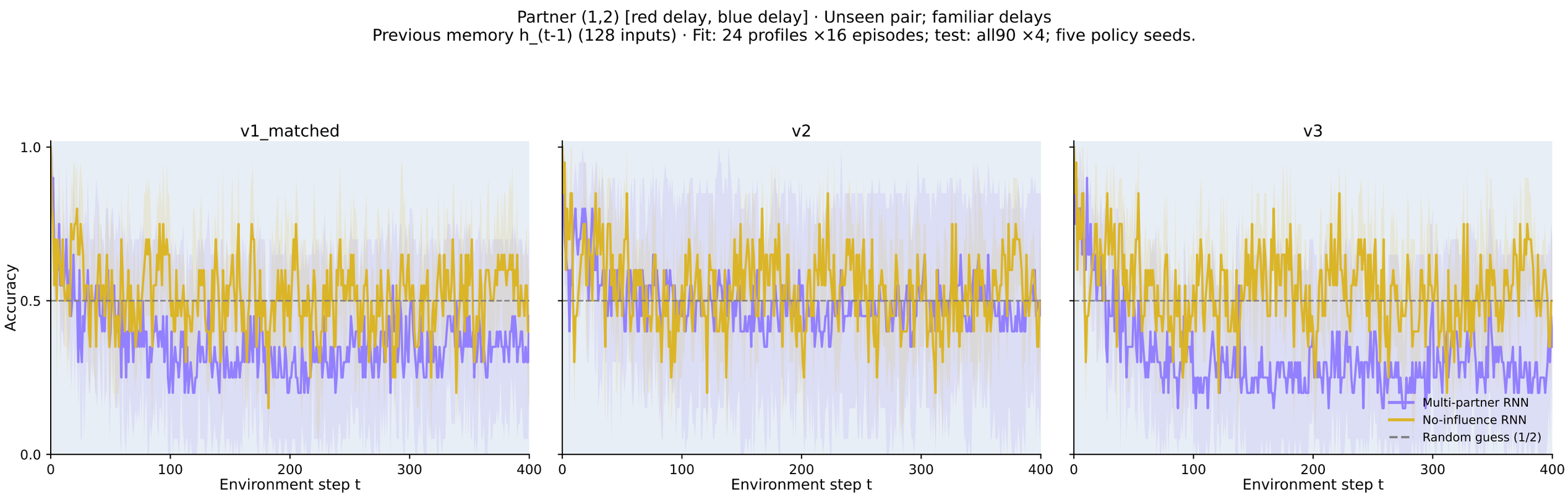

Partner (0,6) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

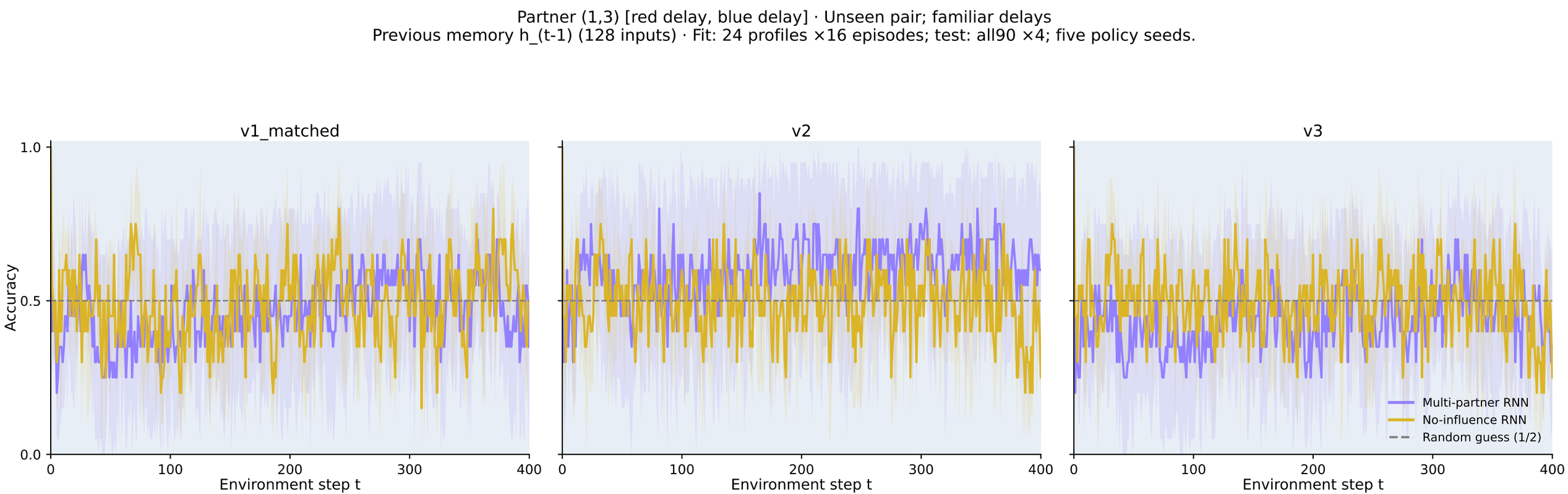

Partner (0,7) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (0,8) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

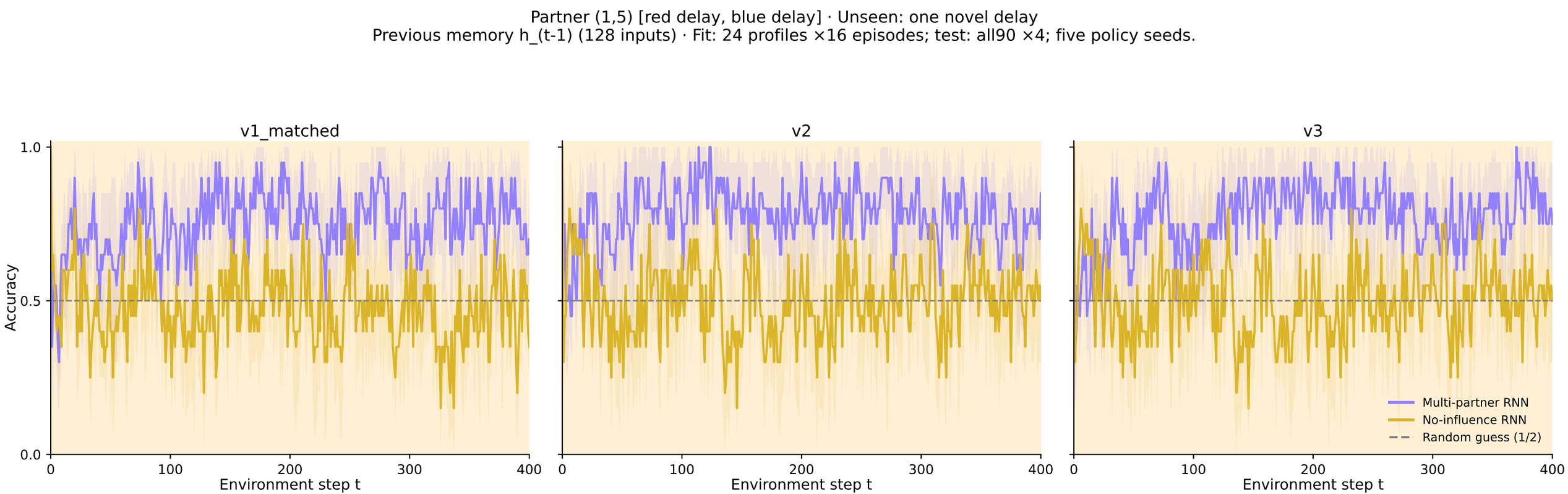

Partner (0,9) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

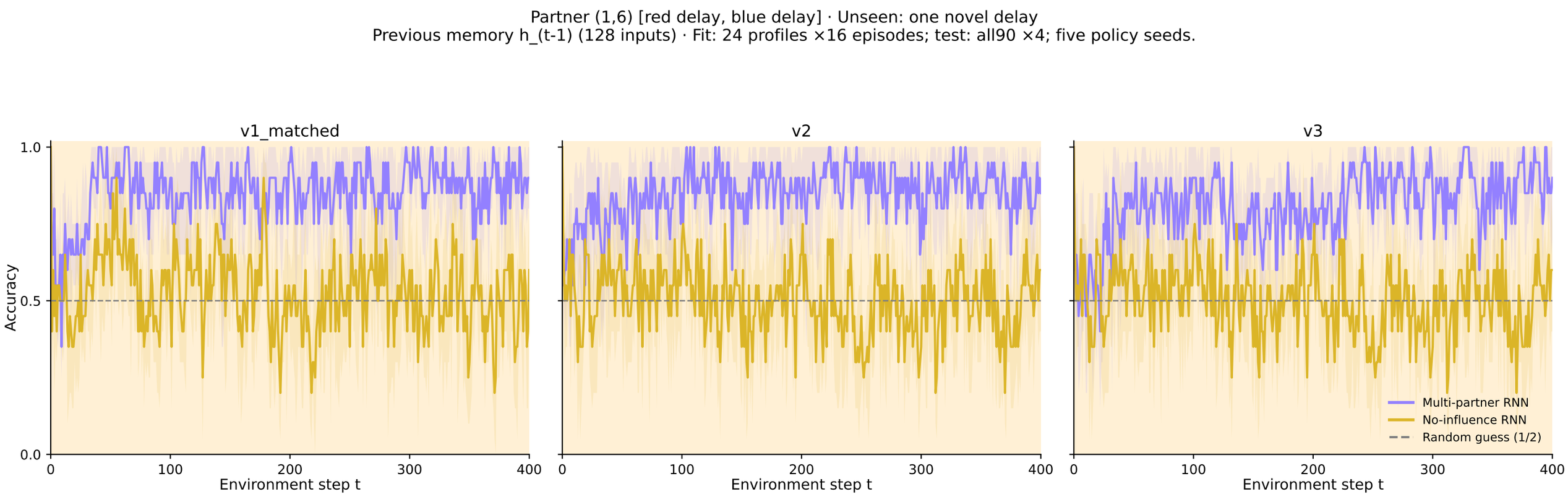

Partner (1,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

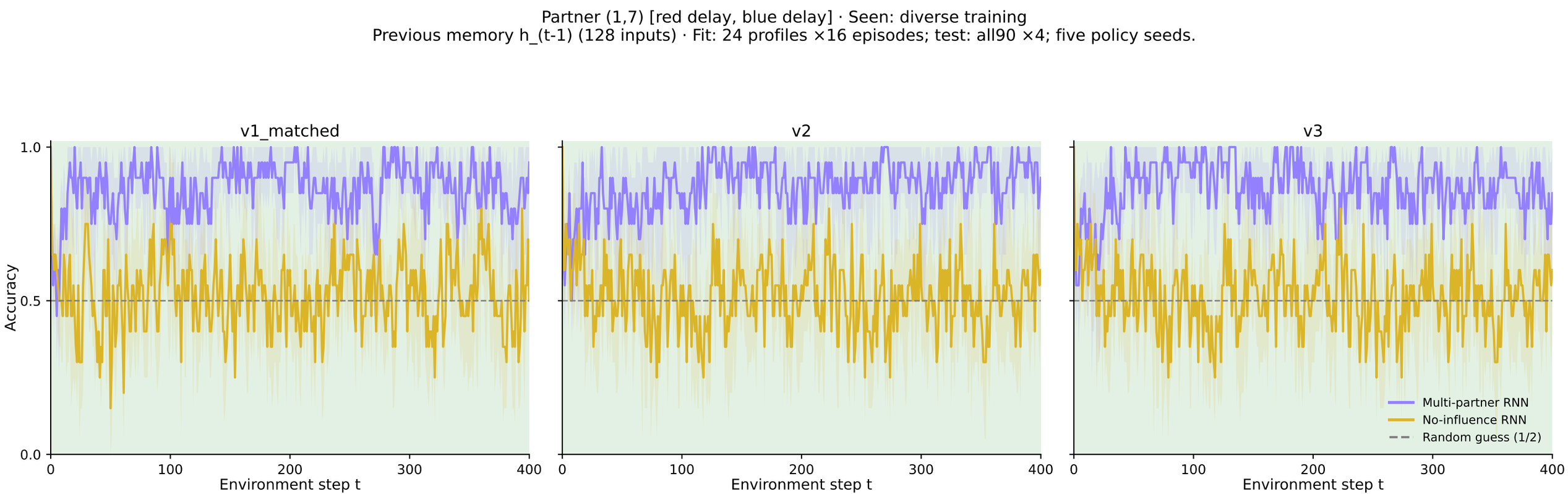

Partner (1,2) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

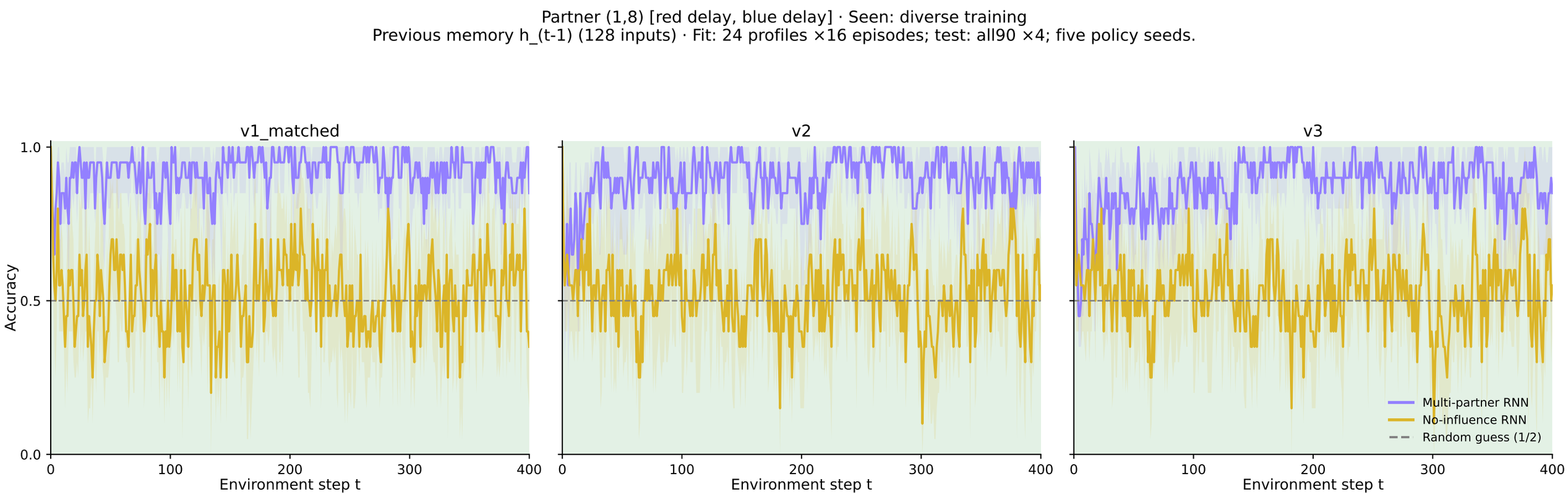

Partner (1,3) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

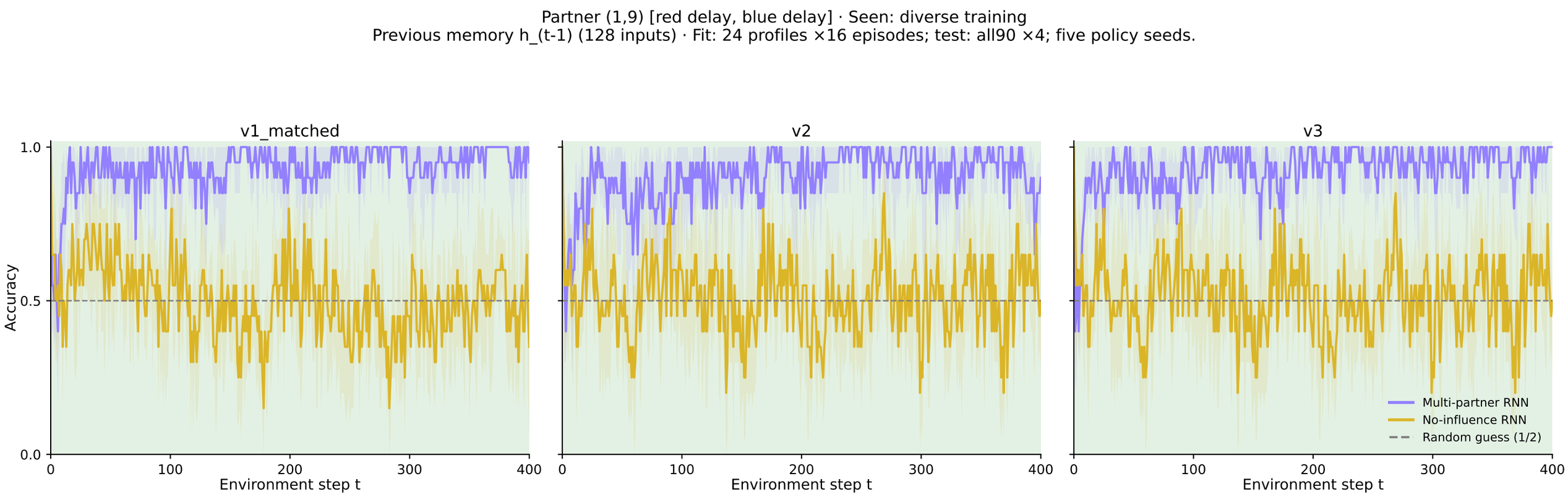

Partner (1,4) [red delay, blue delay] · Seen: diverse training; also single-partner training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

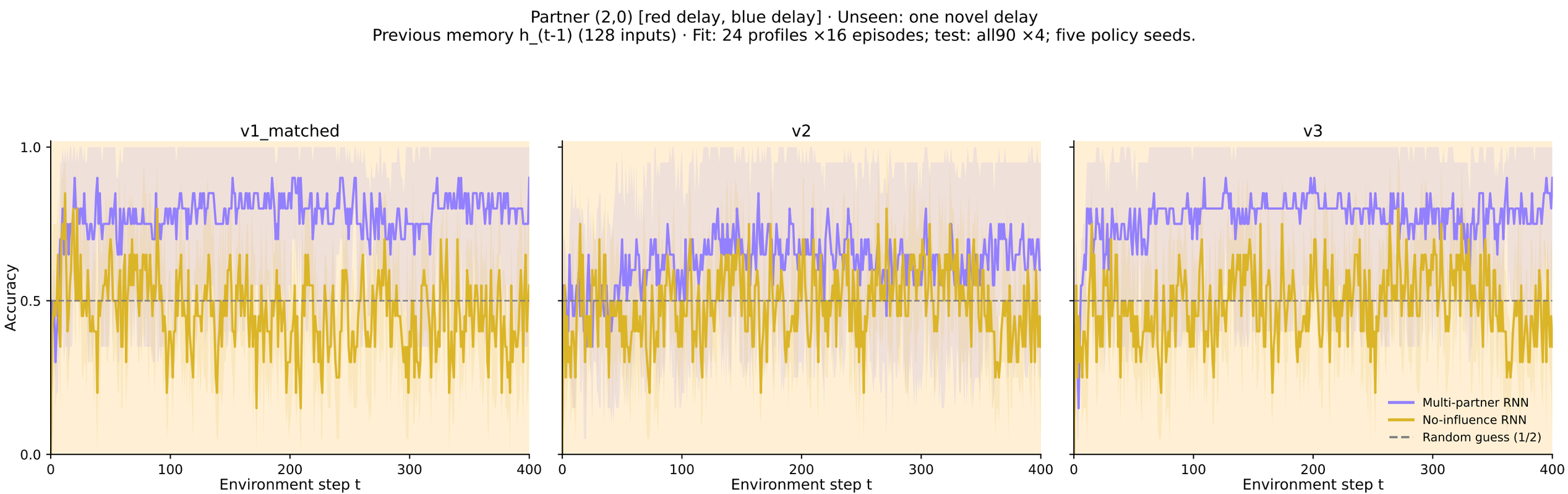

Partner (1,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

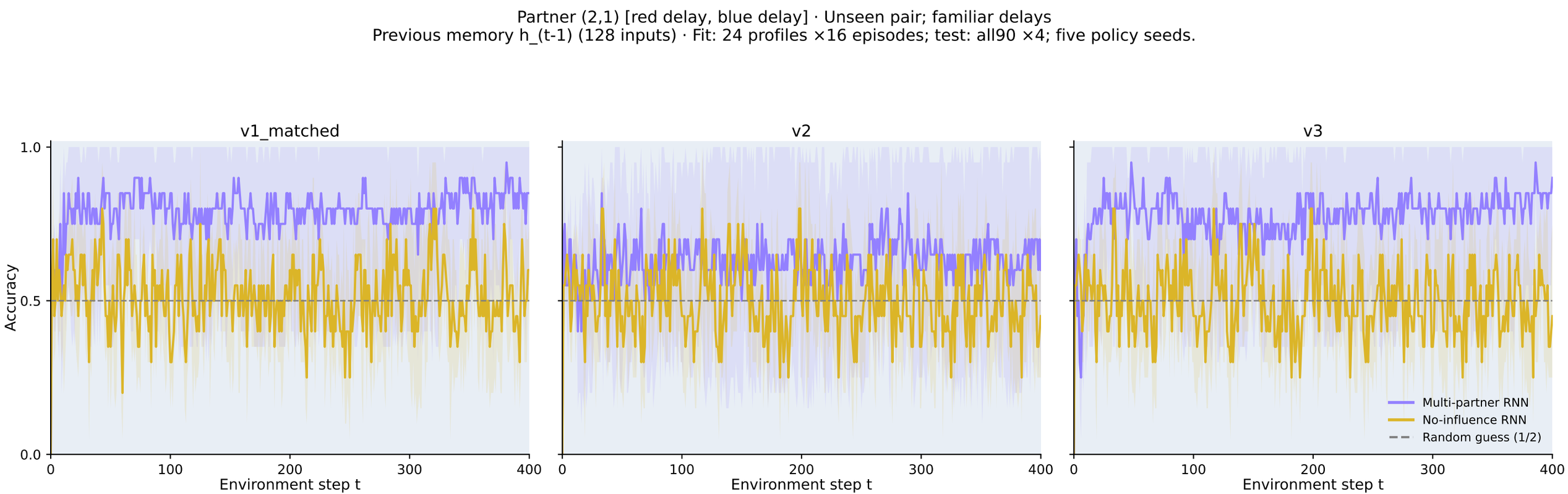

Partner (1,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

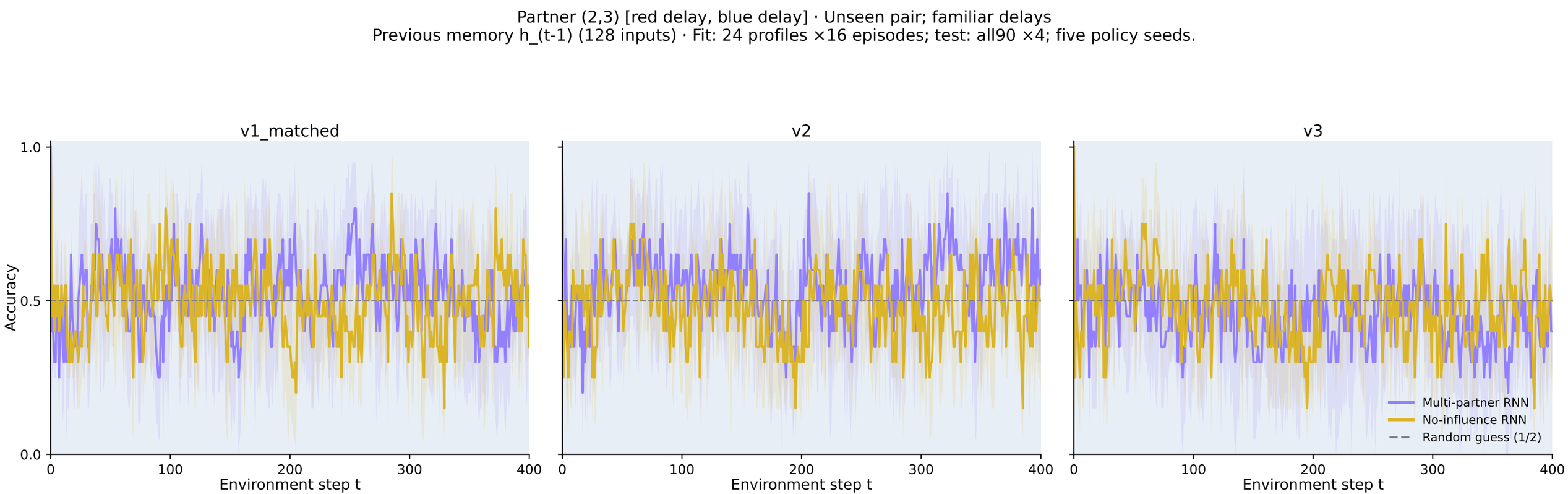

Partner (1,7) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

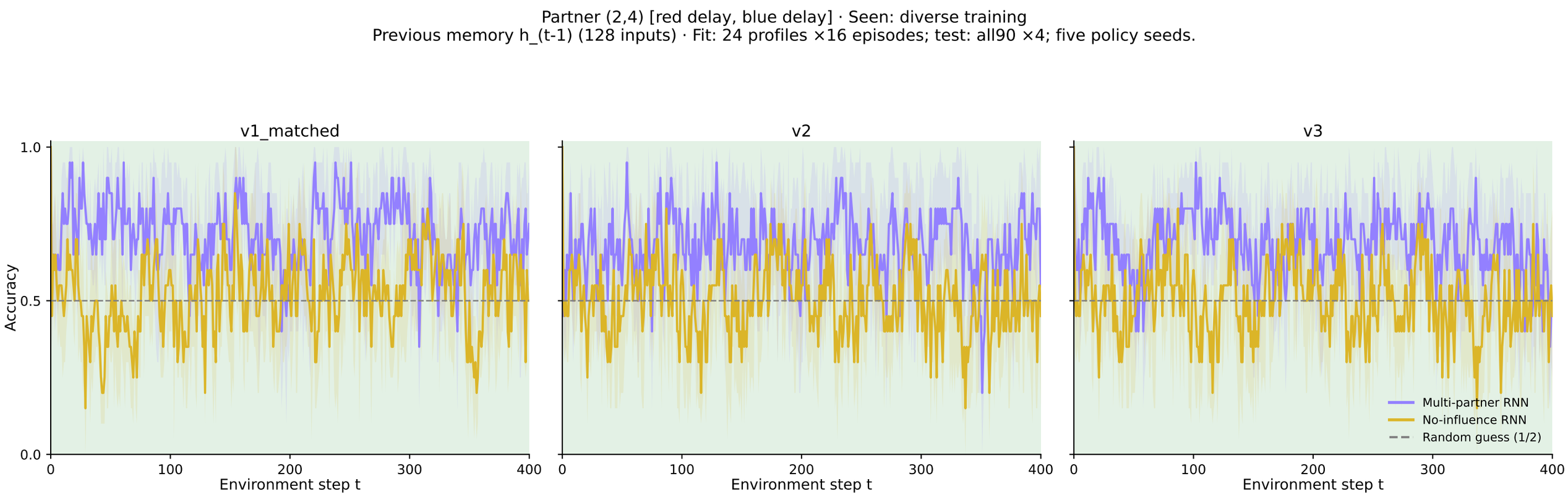

Partner (1,8) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

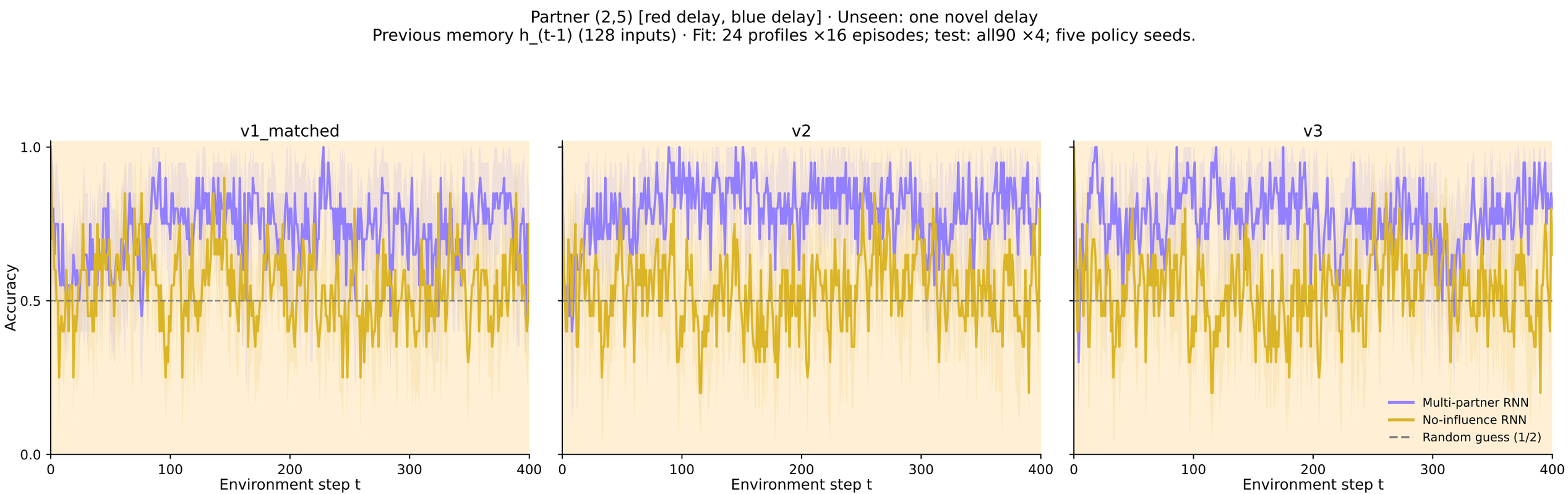

Partner (1,9) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

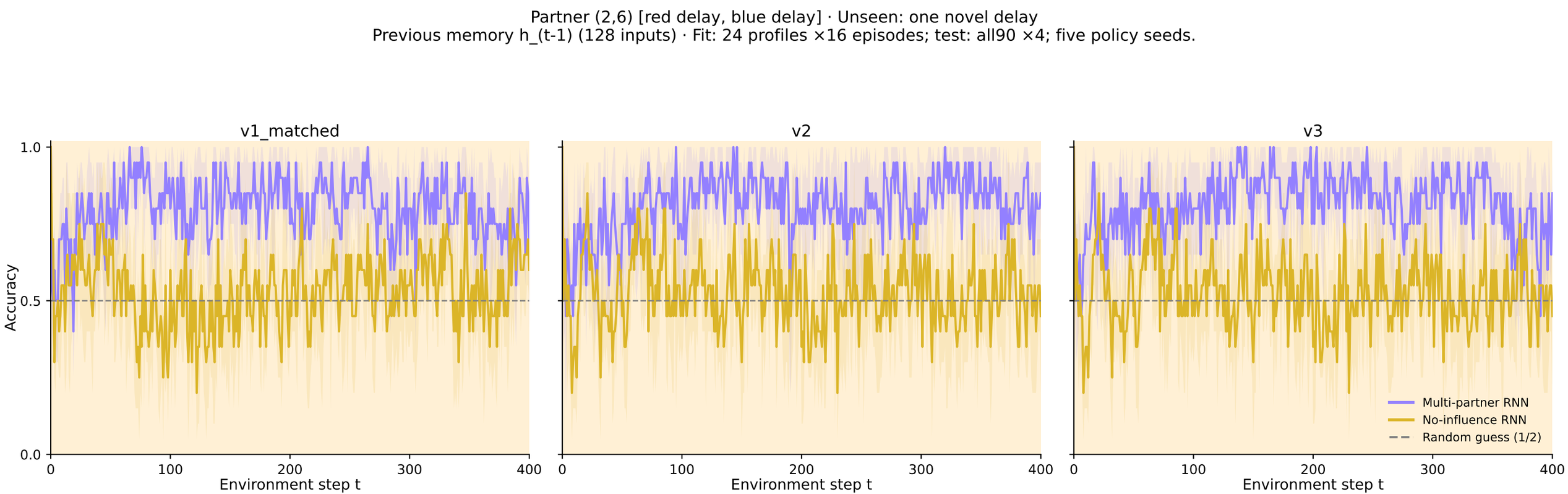

Partner (2,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

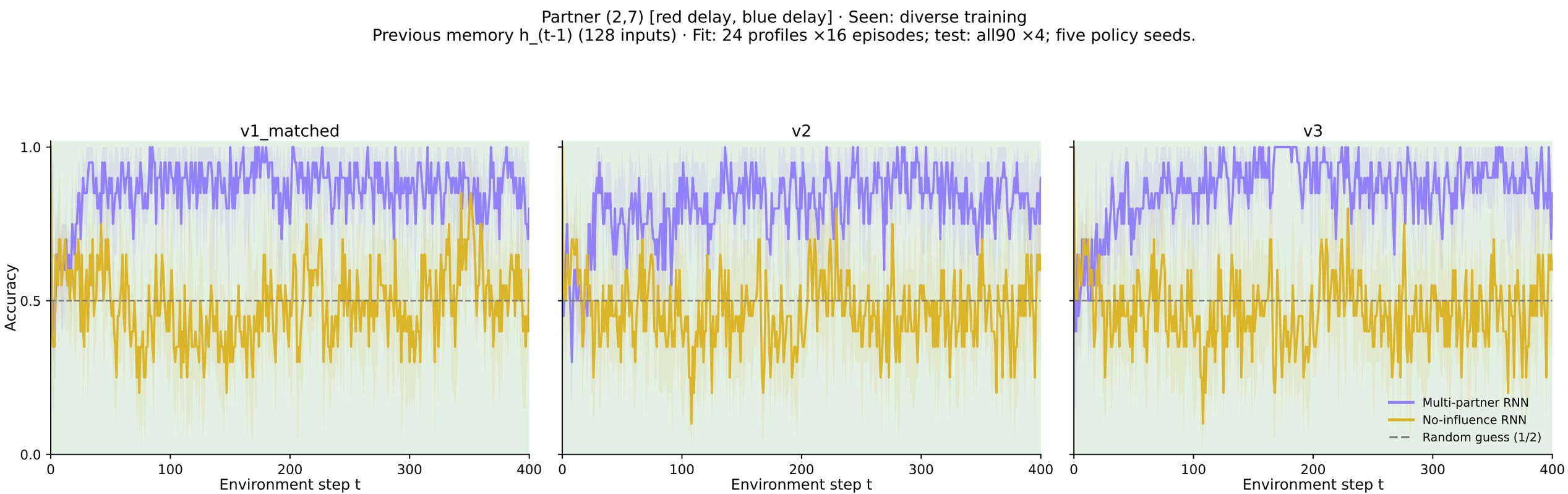

Partner (2,1) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

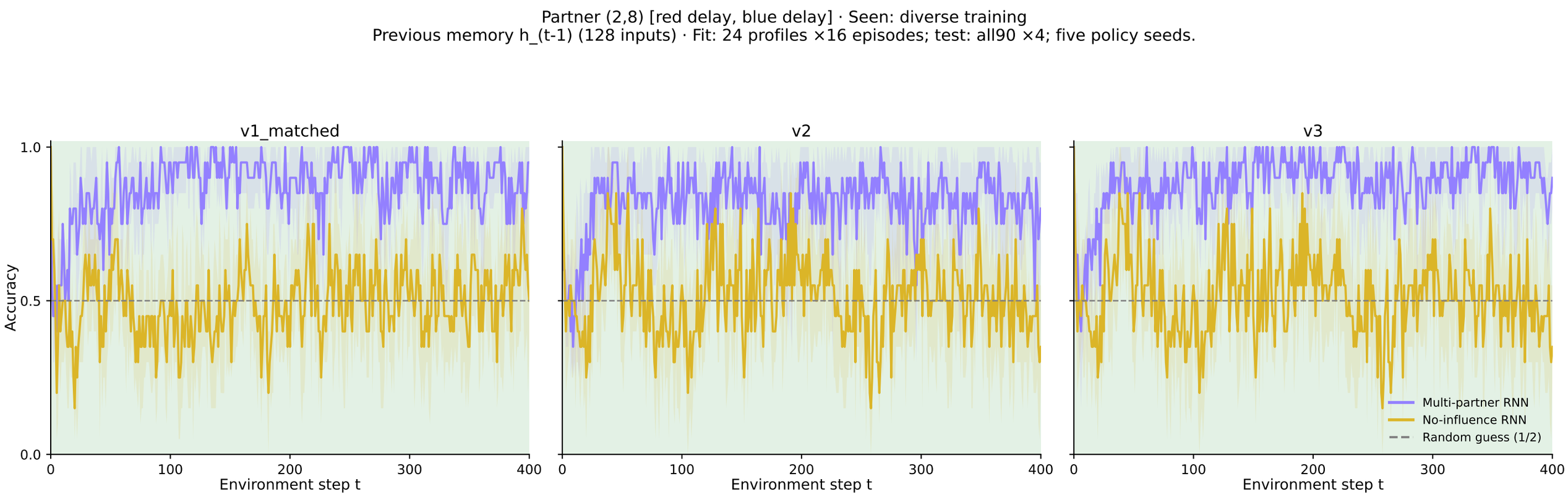

Partner (2,3) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

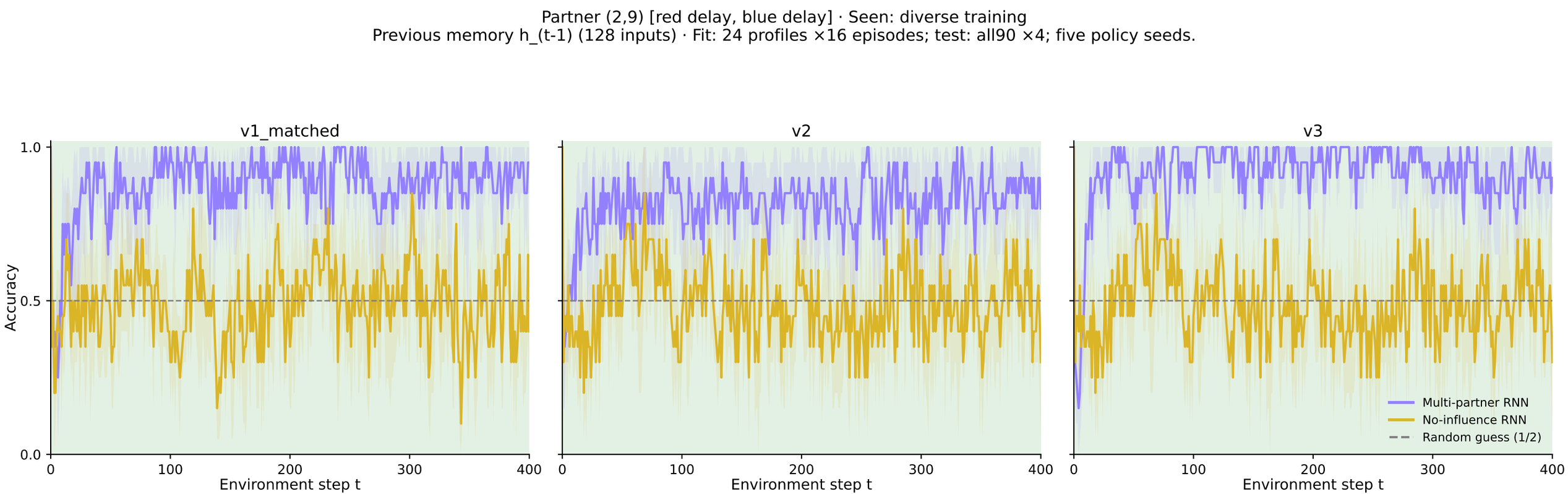

Partner (2,4) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

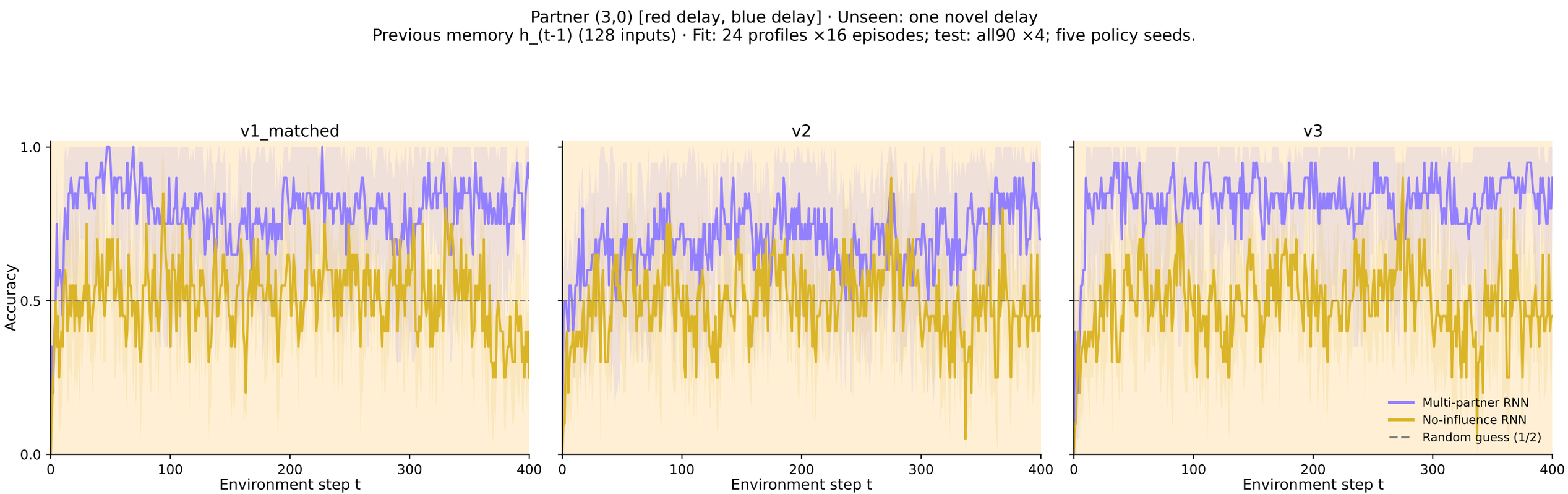

Partner (2,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

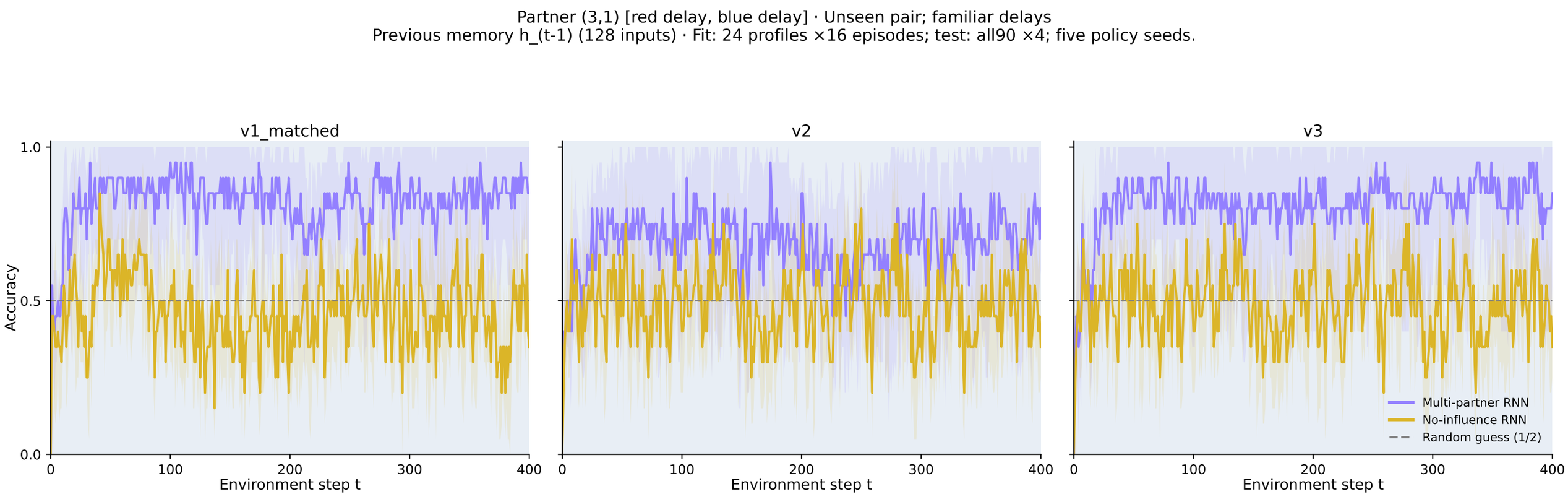

Partner (2,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

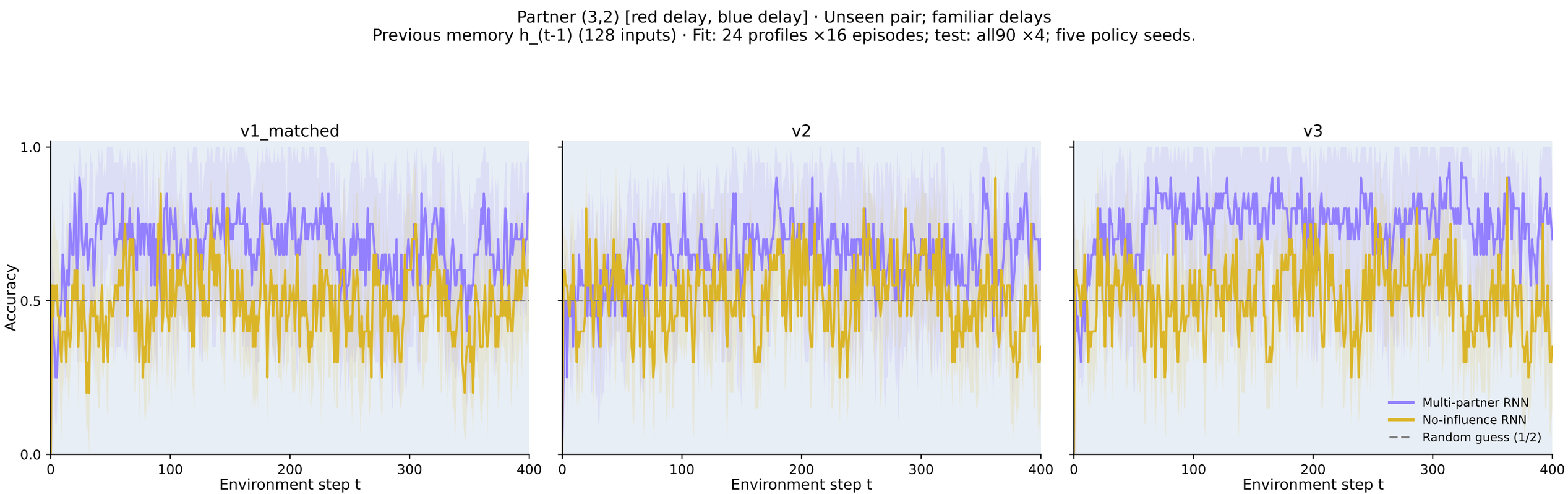

Partner (2,7) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

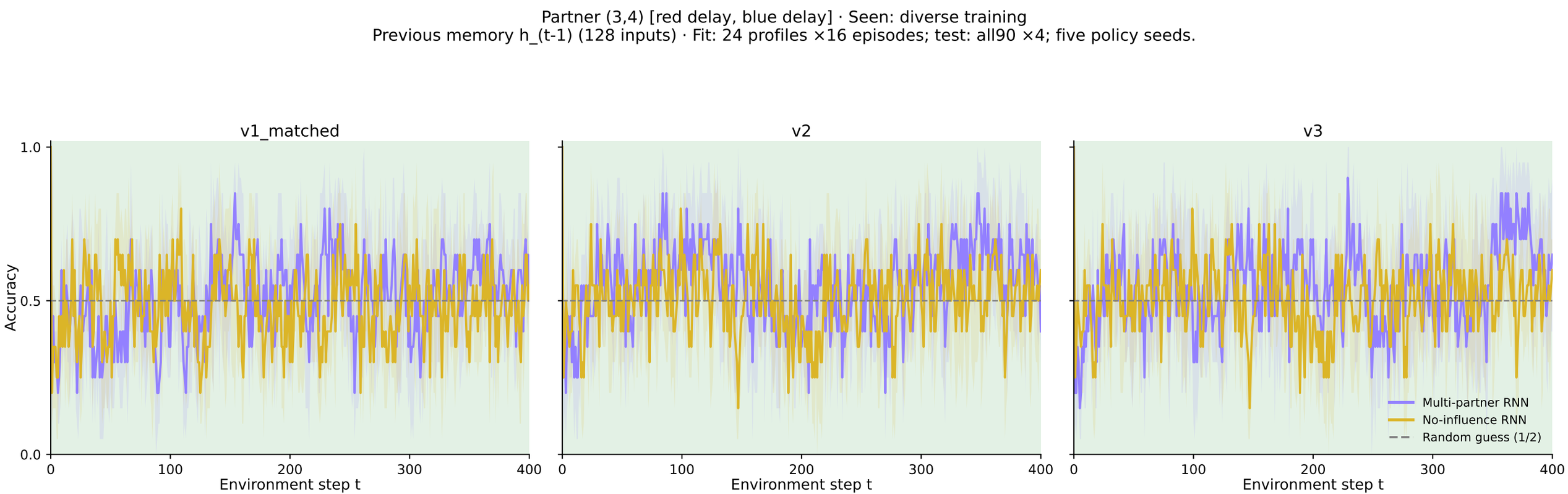

Partner (2,8) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

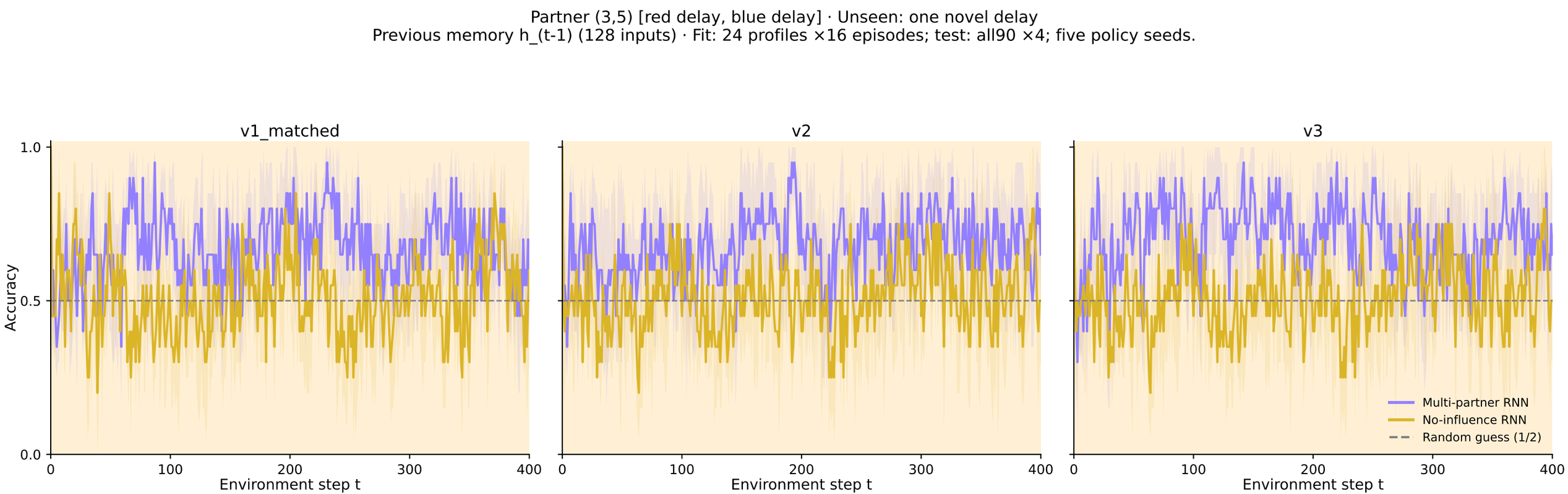

Partner (2,9) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

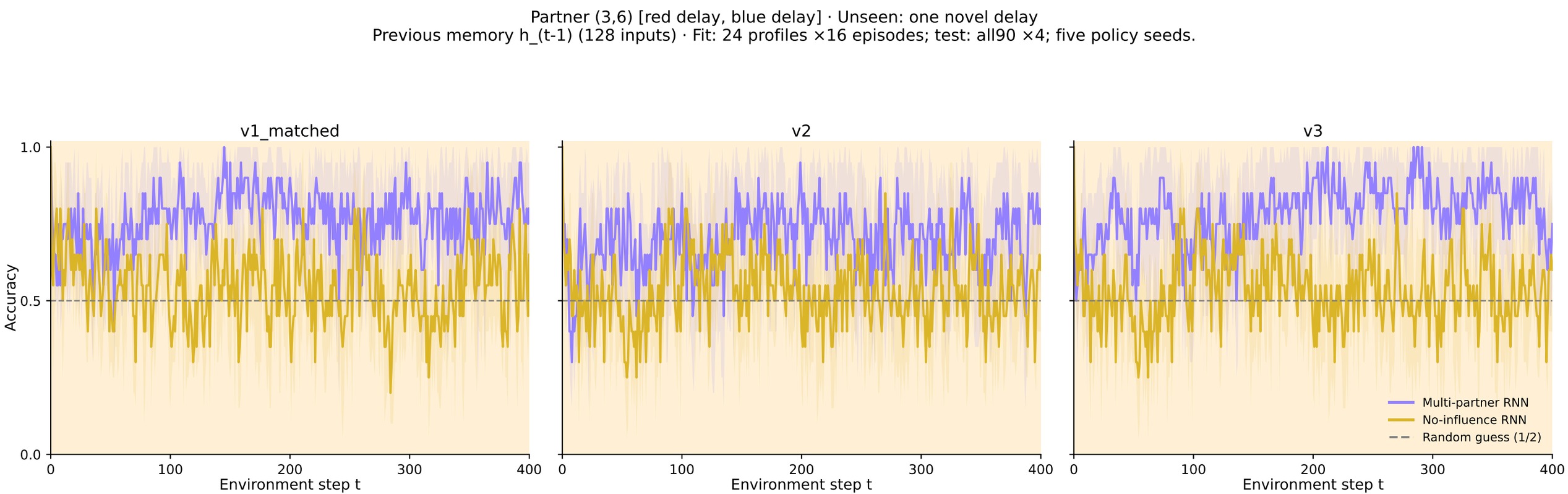

Partner (3,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

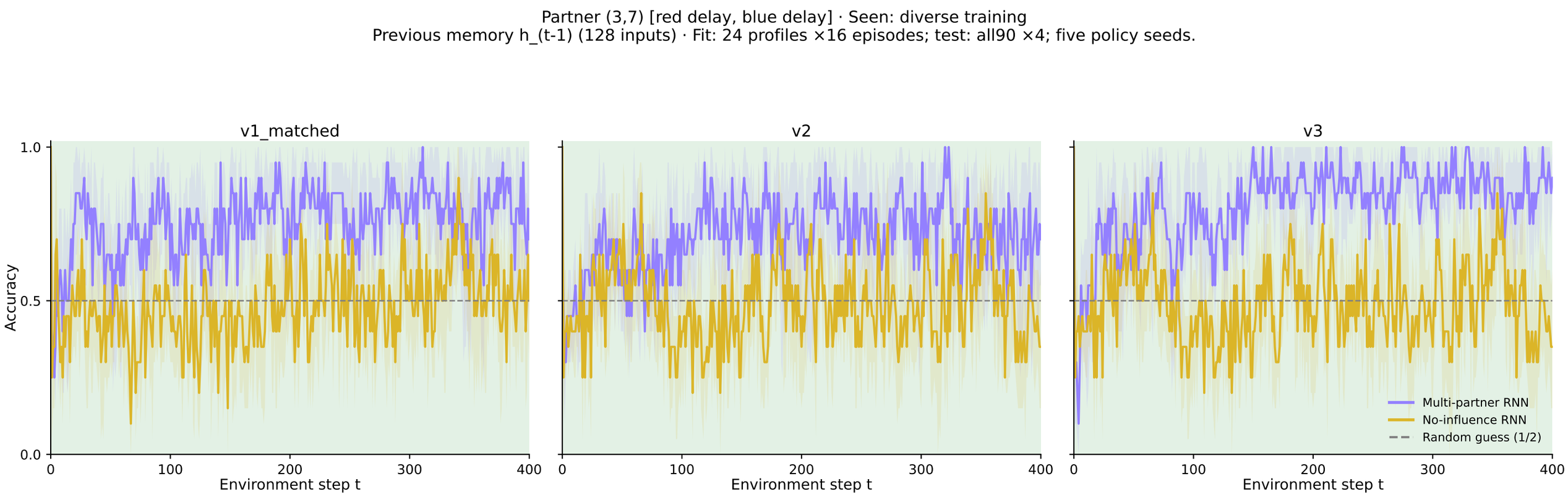

Partner (3,1) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

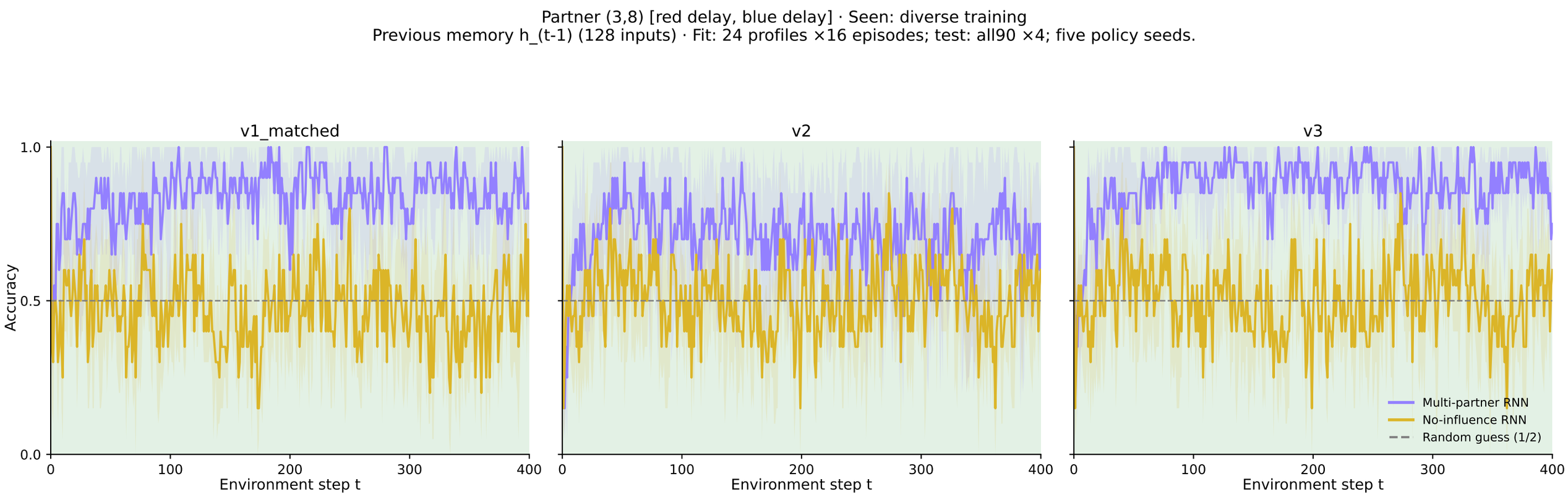

Partner (3,2) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

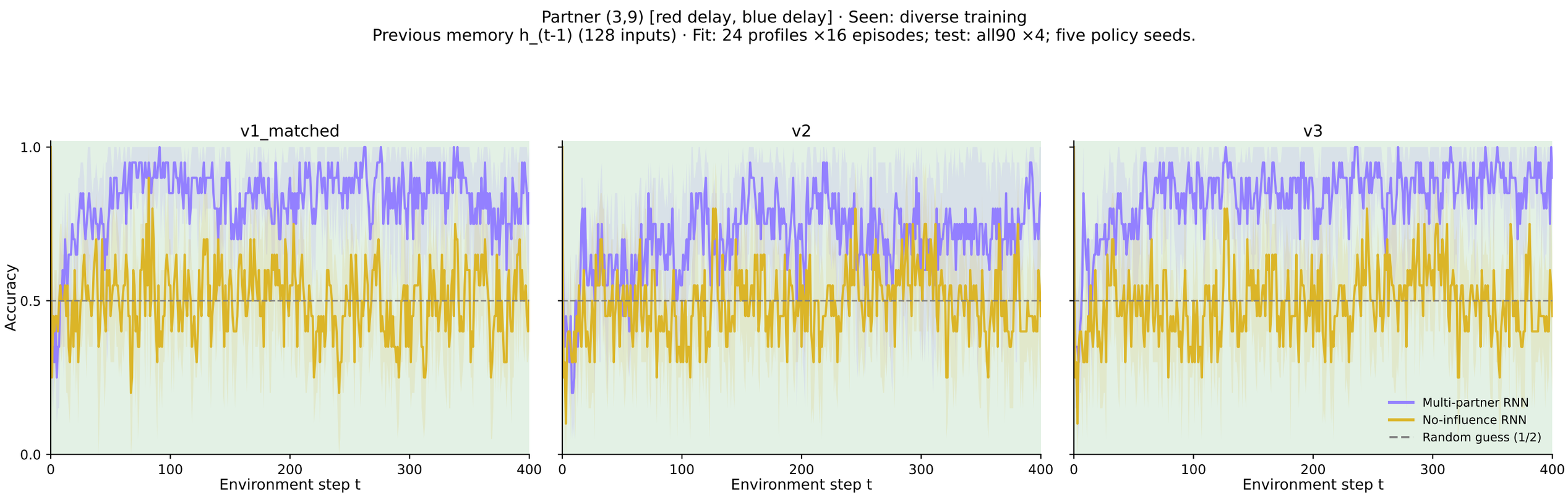

Partner (3,4) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

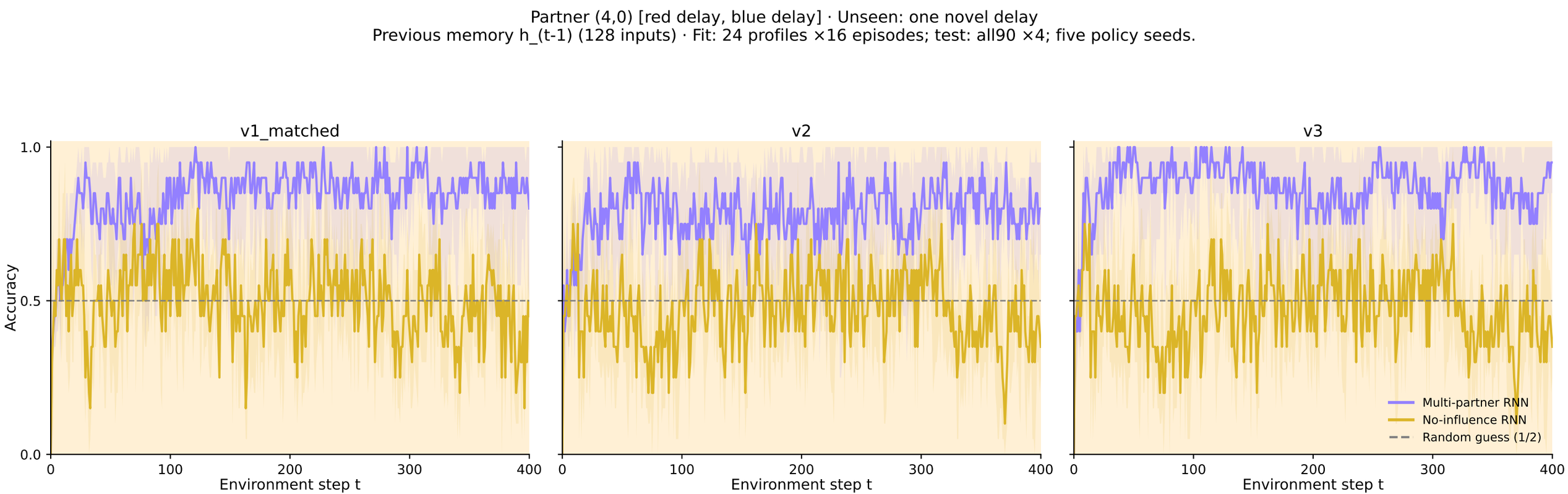

Partner (3,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

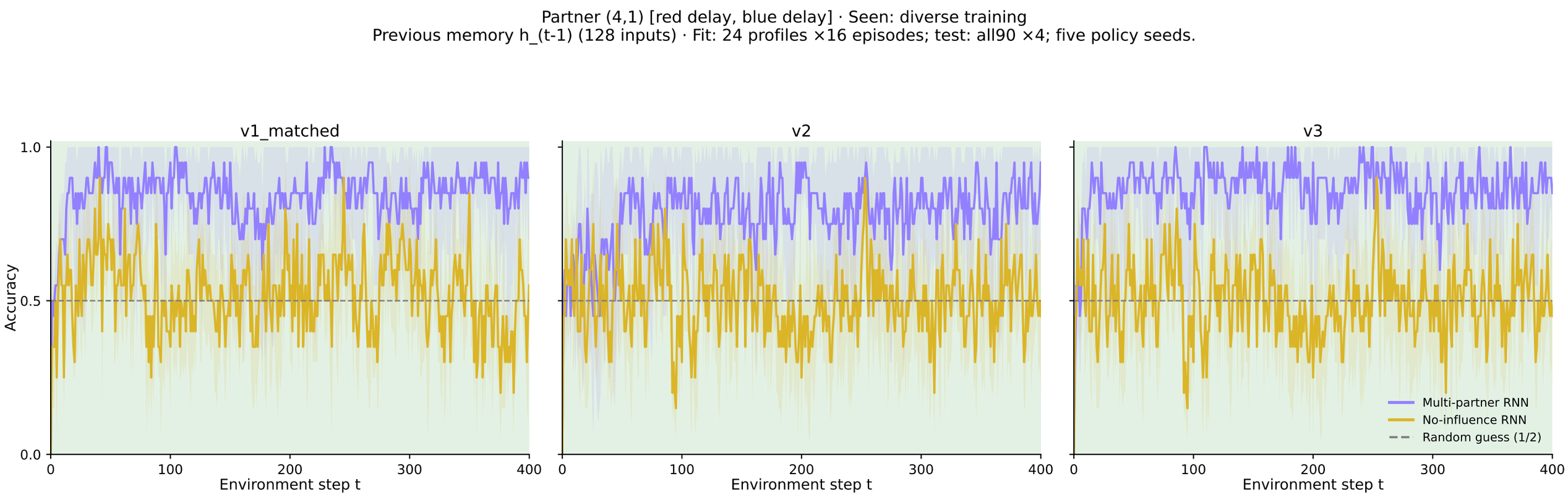

Partner (3,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

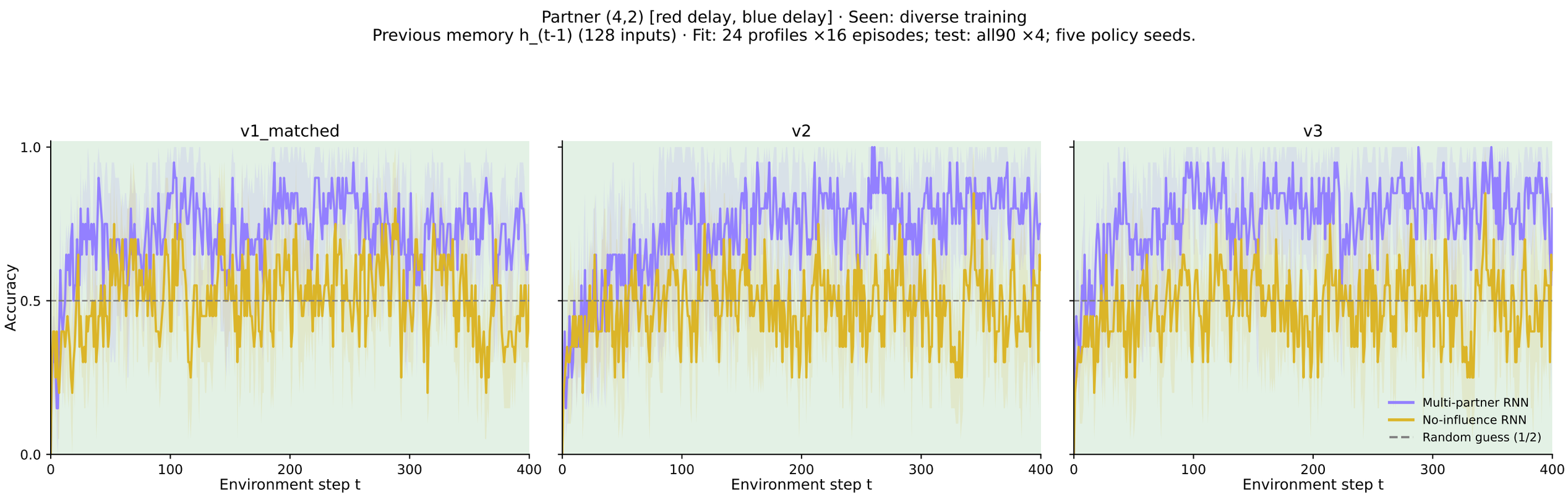

Partner (3,7) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

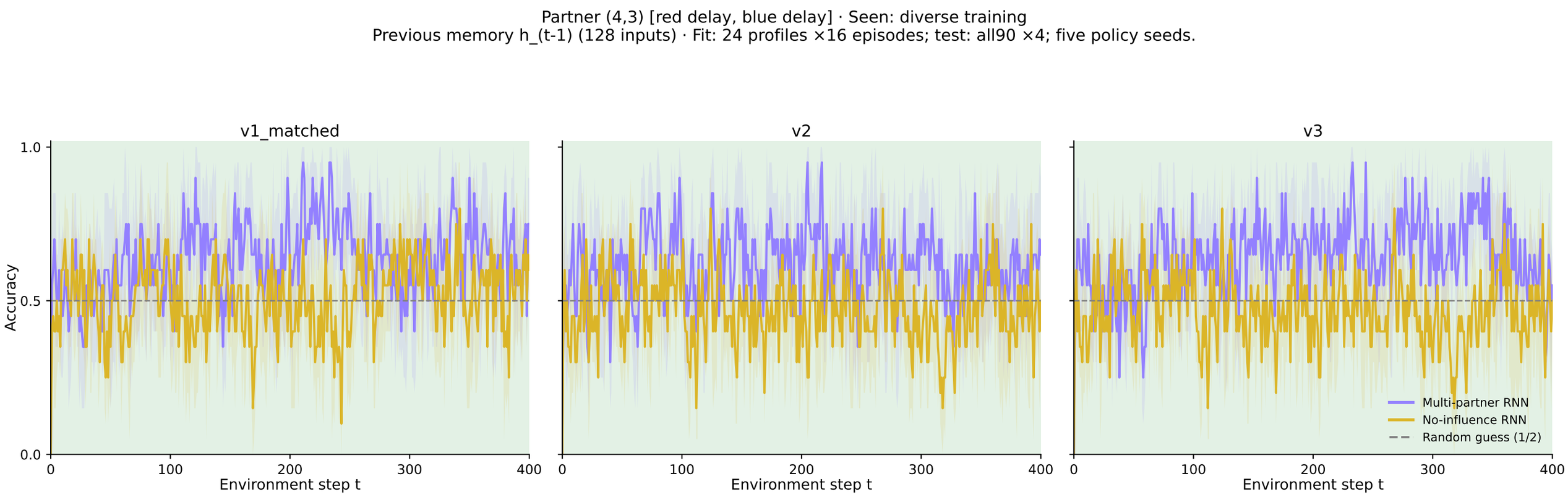

Partner (3,8) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

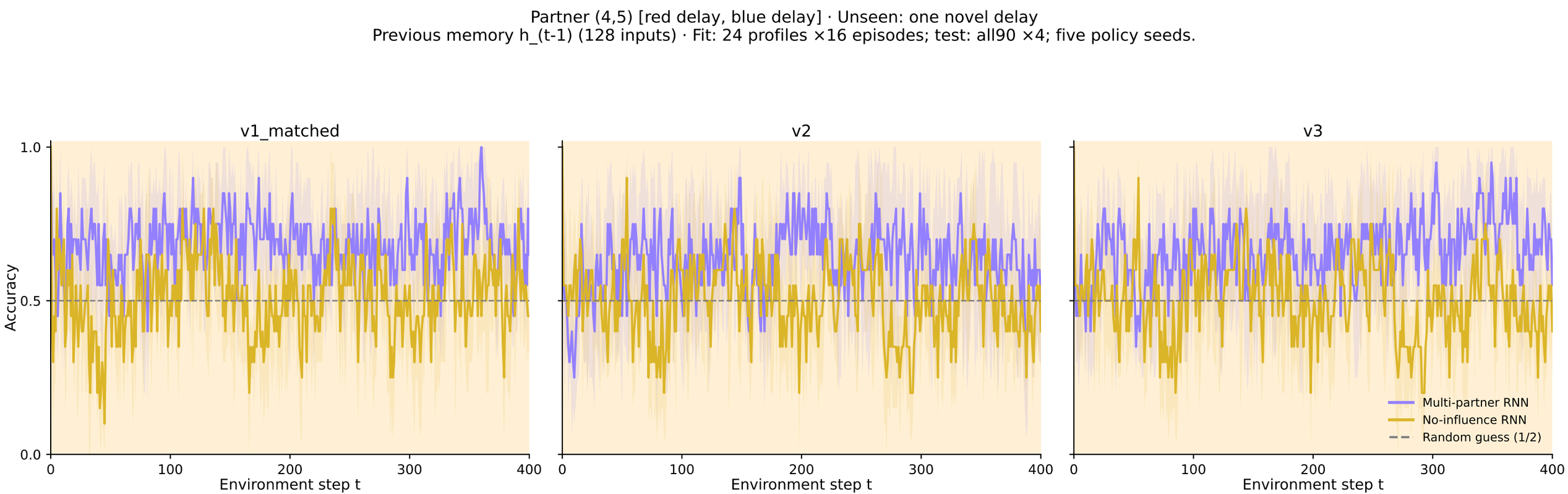

Partner (3,9) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

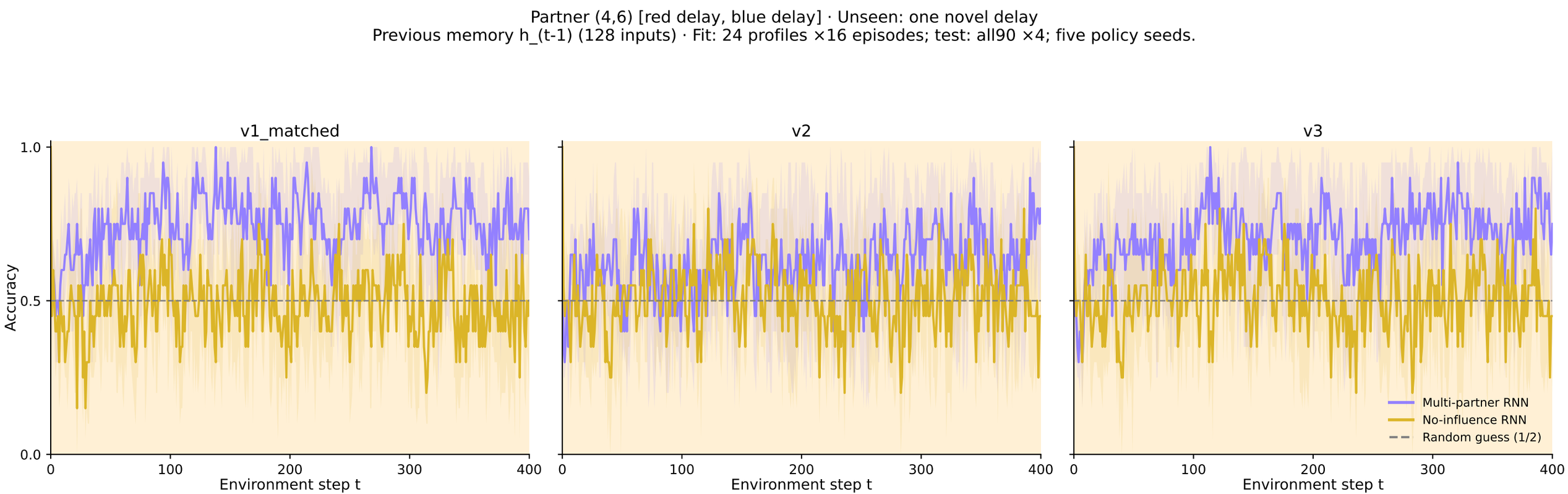

Partner (4,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

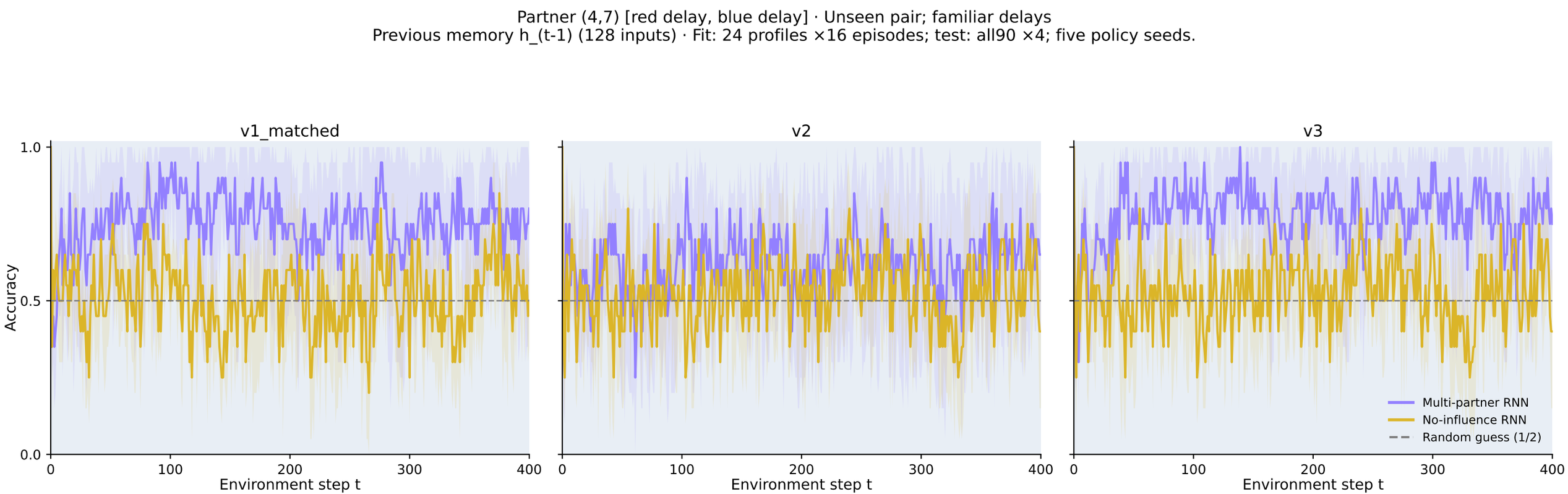

Partner (4,1) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

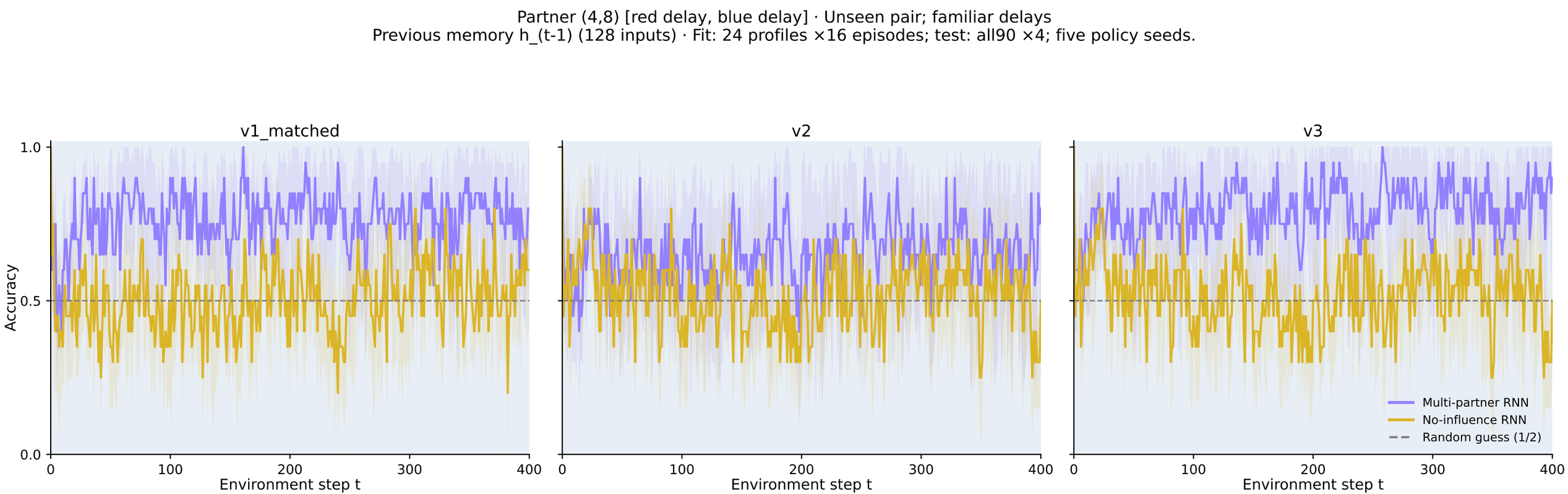

Partner (4,2) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

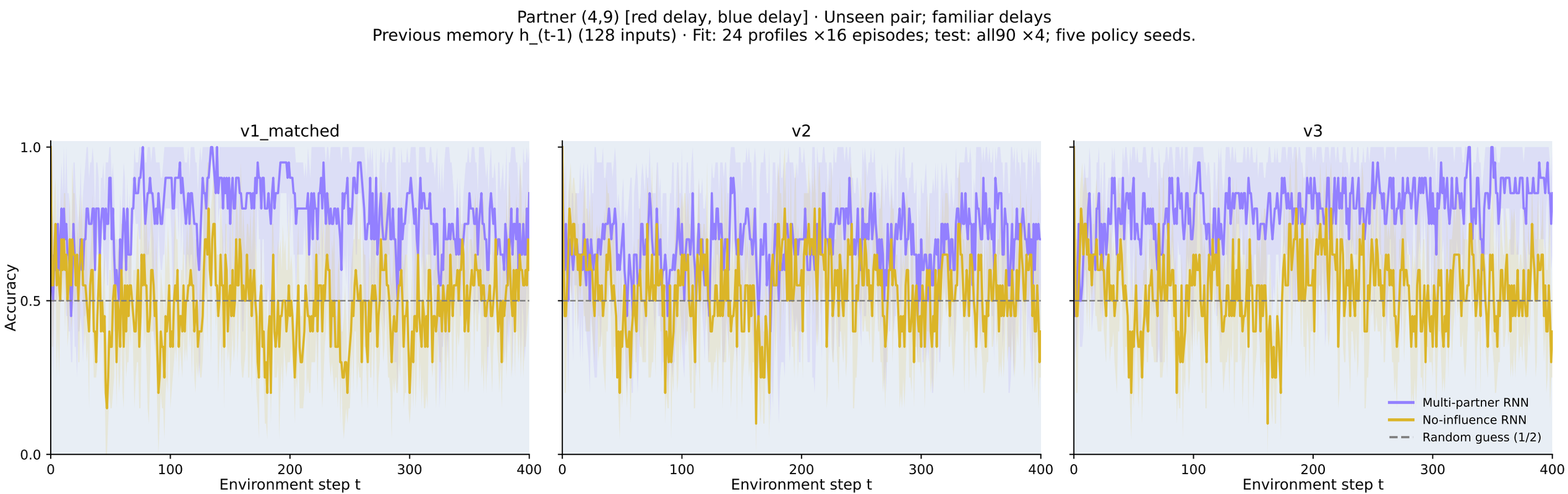

Partner (4,3) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

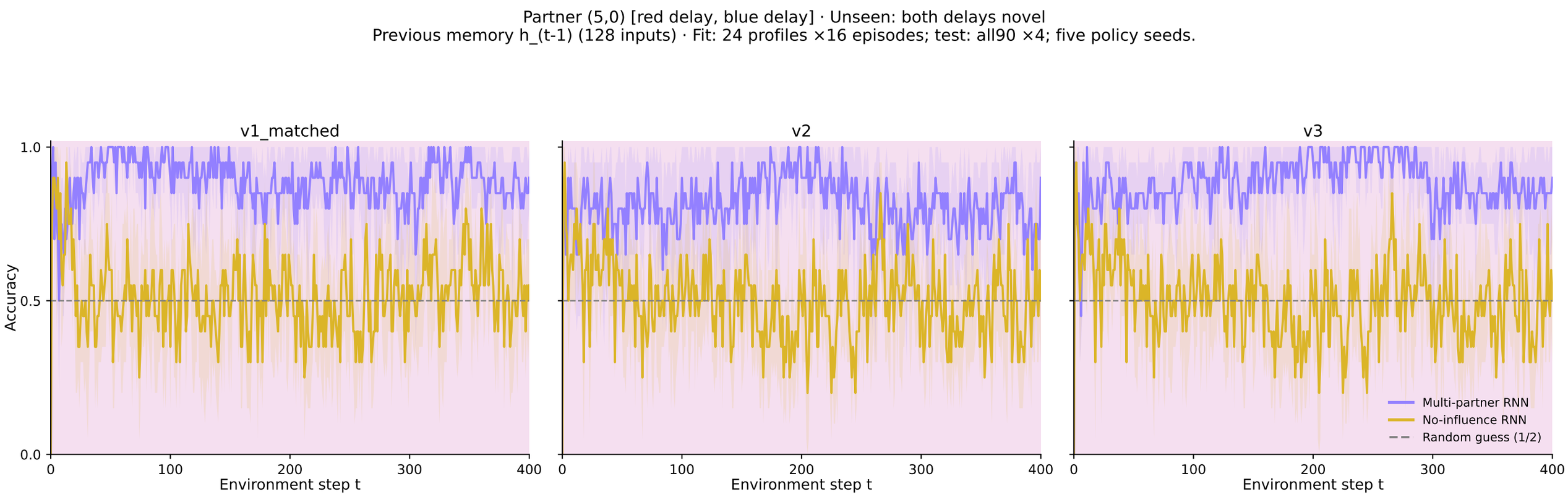

Partner (4,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

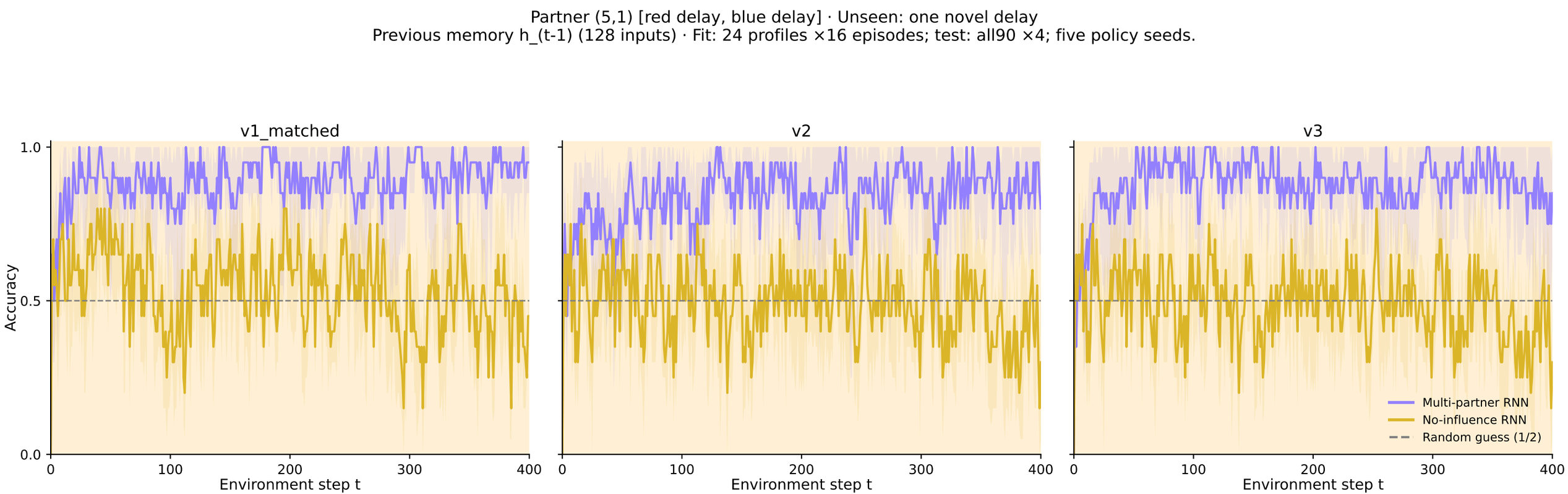

Partner (4,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

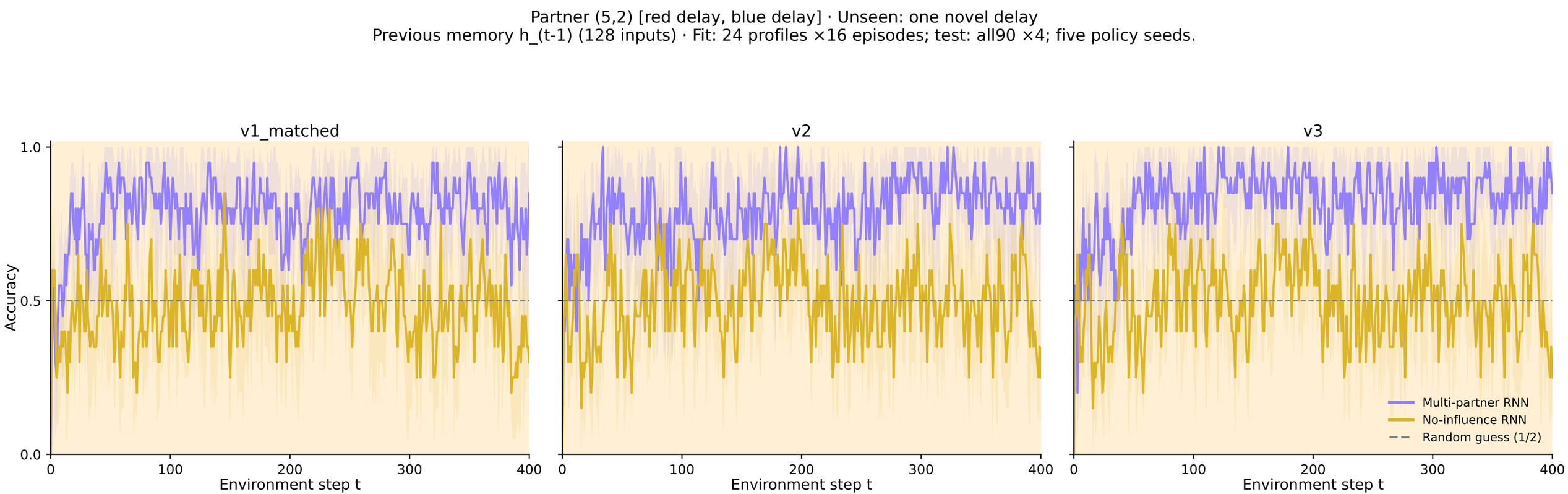

Partner (4,7) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

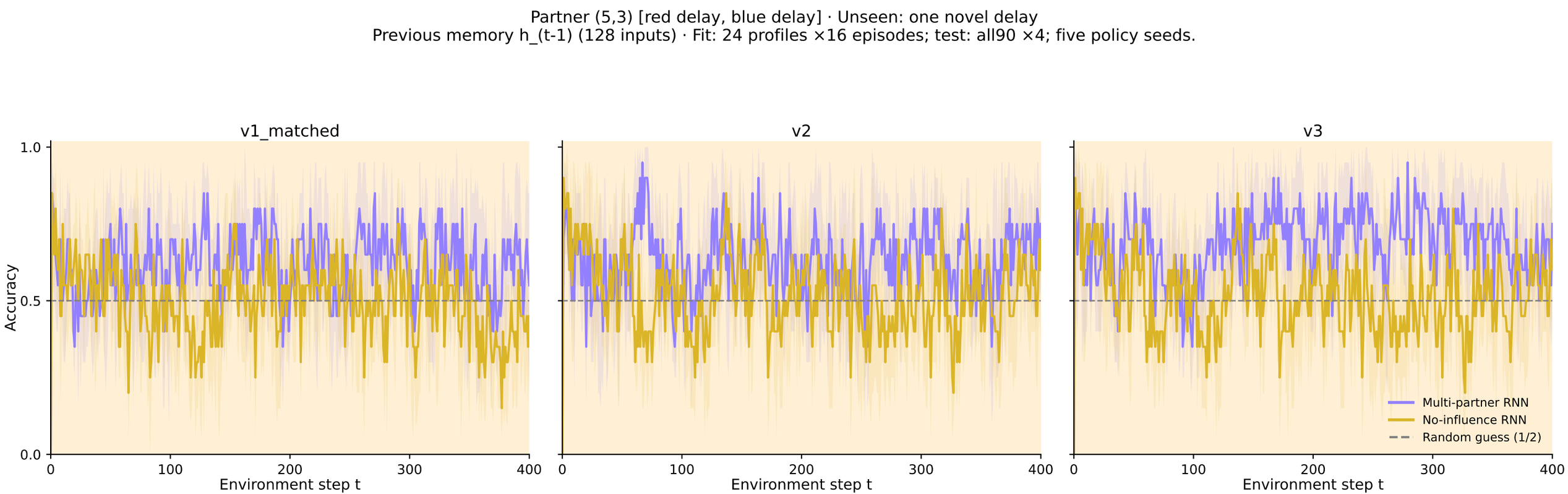

Partner (4,8) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

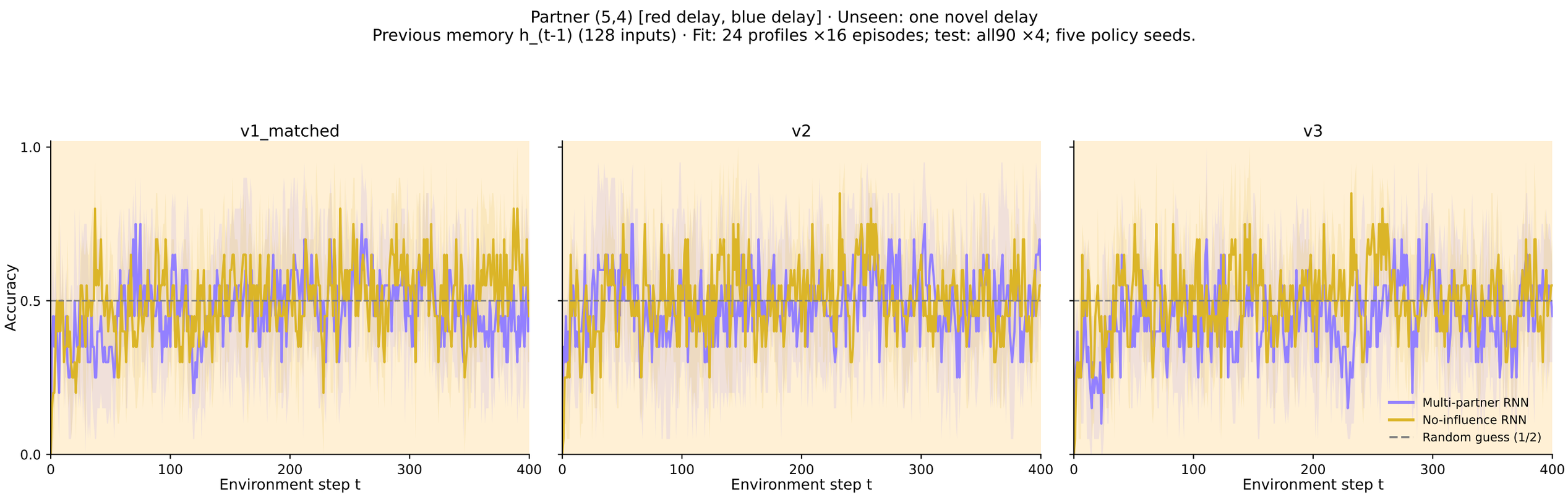

Partner (4,9) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

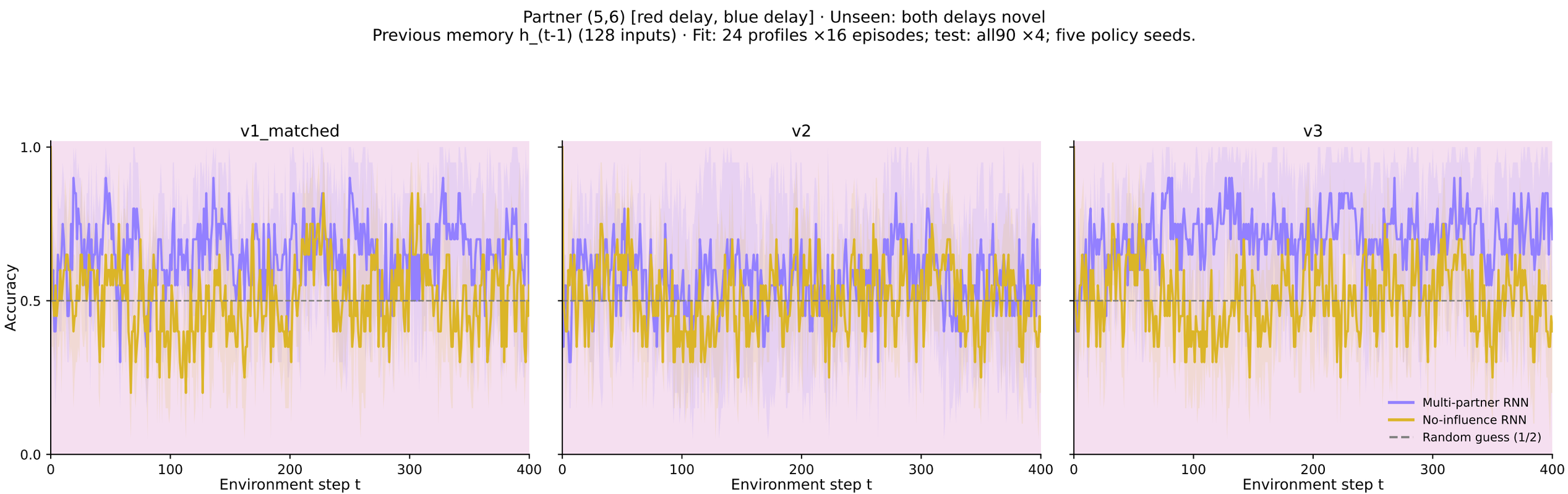

Partner (5,0) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

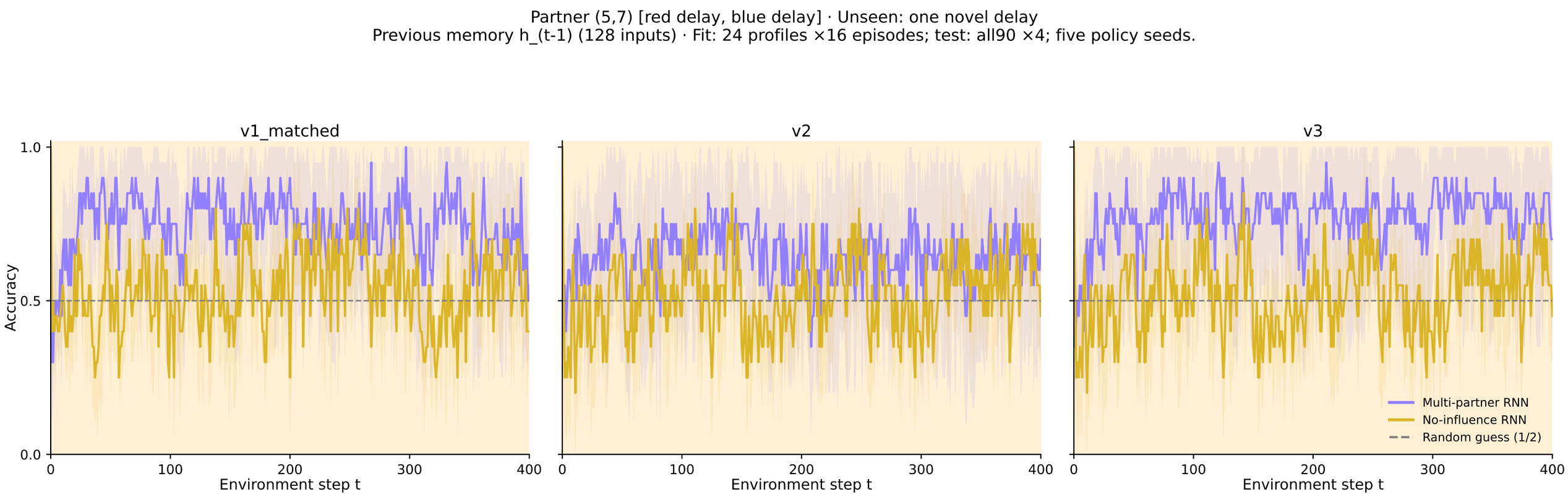

Partner (5,1) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

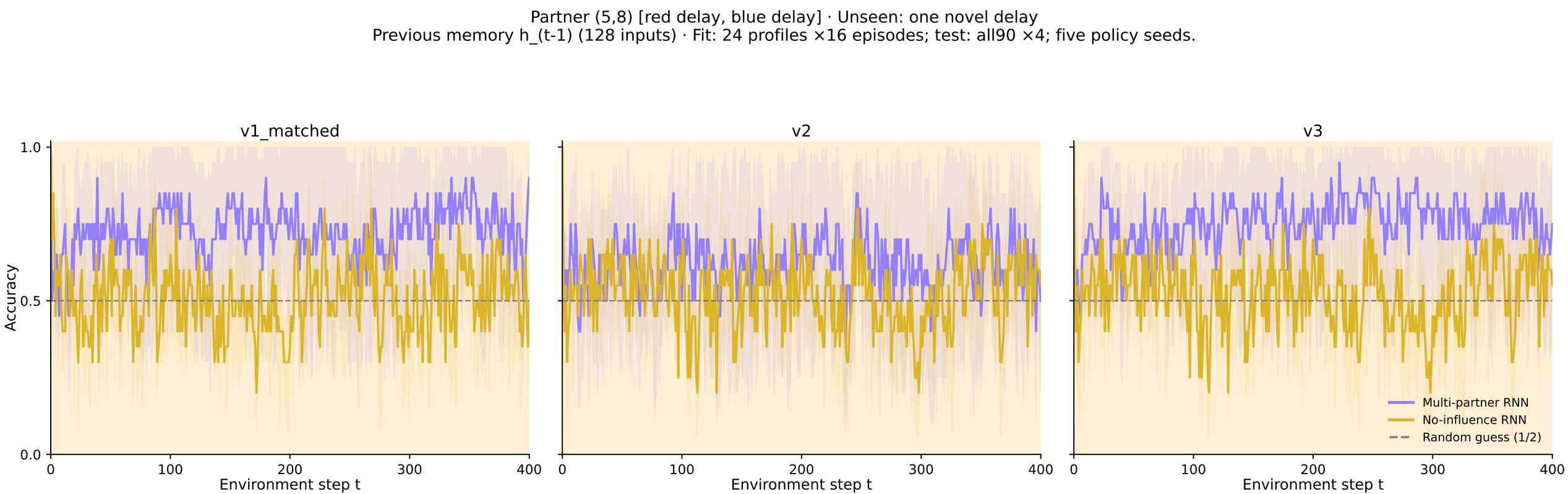

Partner (5,2) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

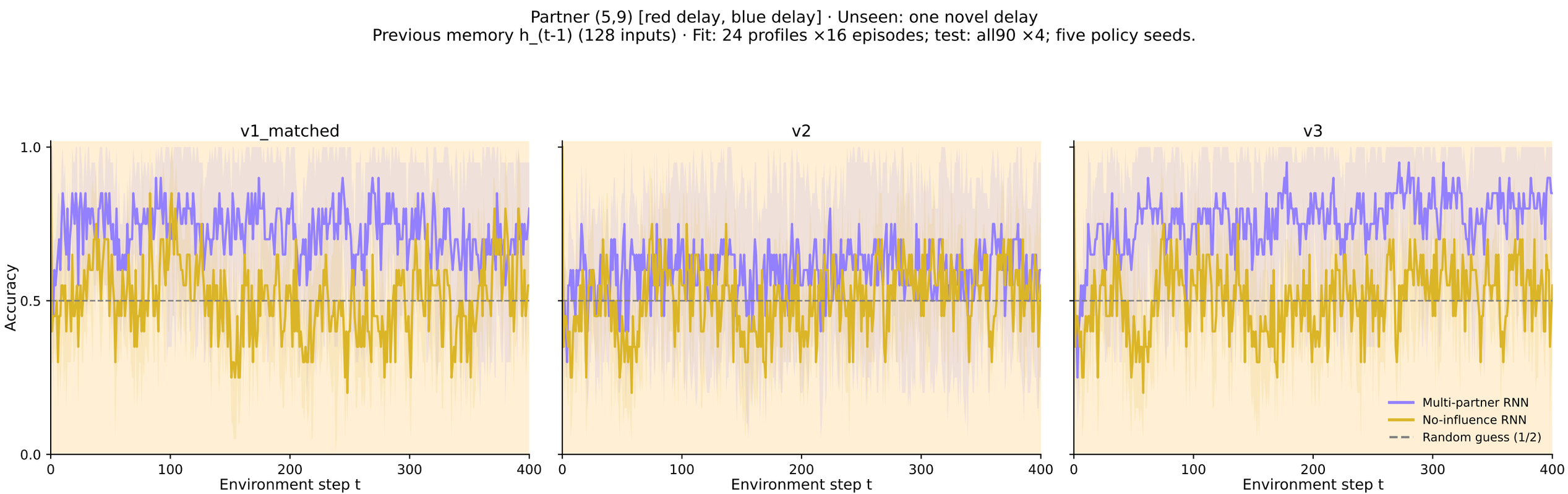

Partner (5,3) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

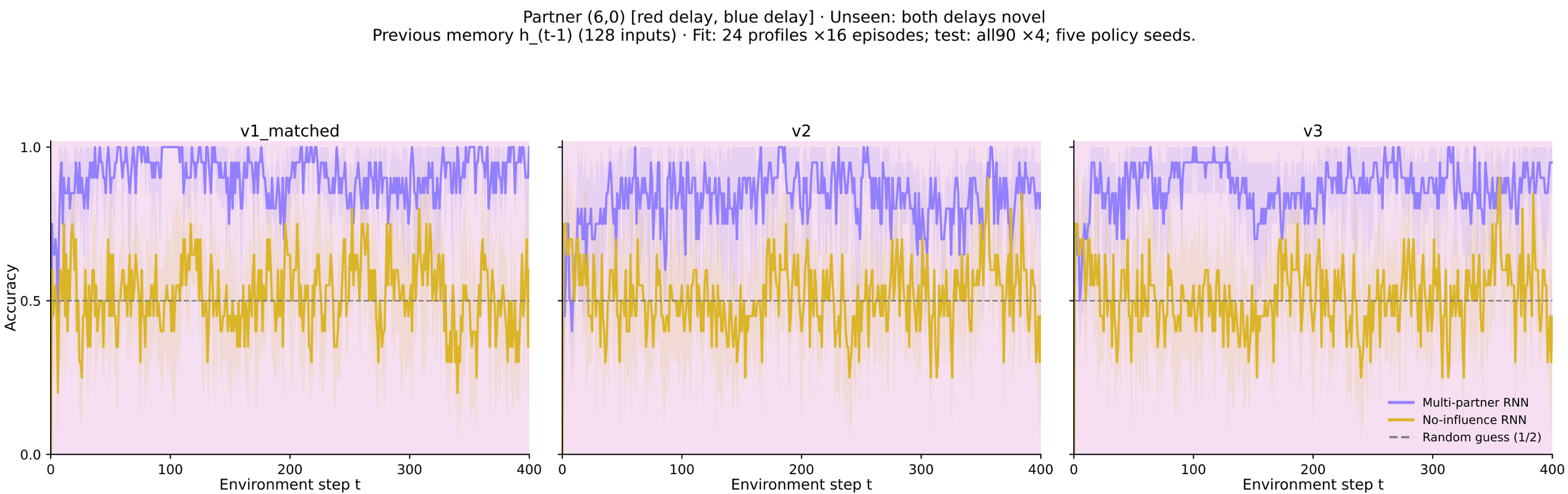

Partner (5,4) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

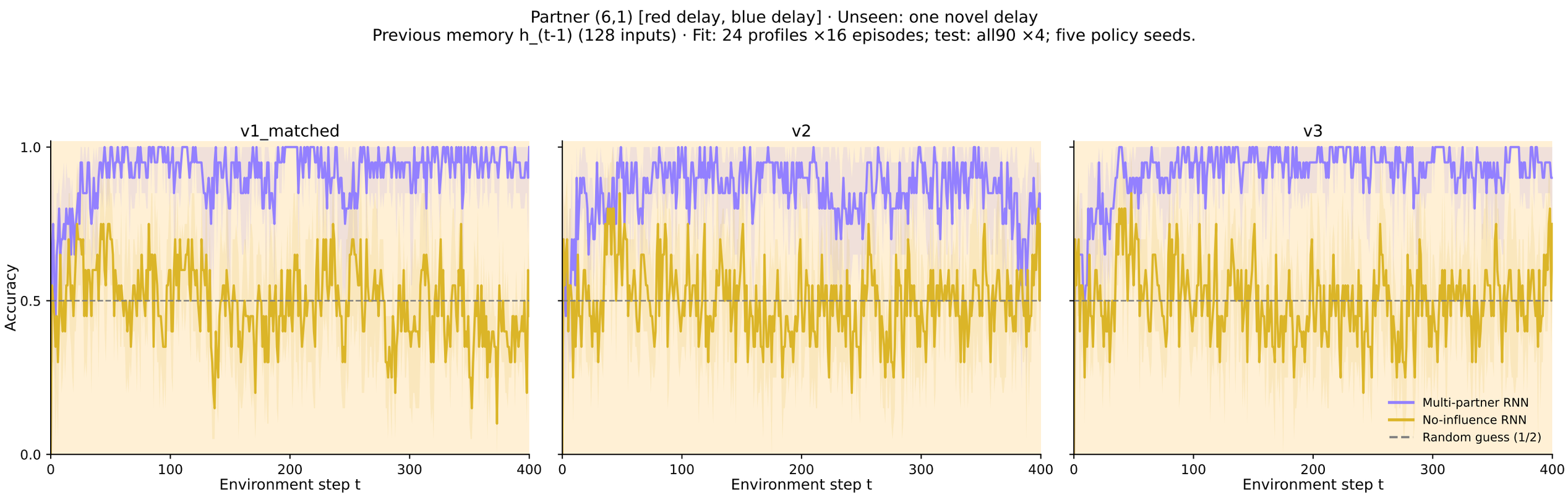

Partner (5,6) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

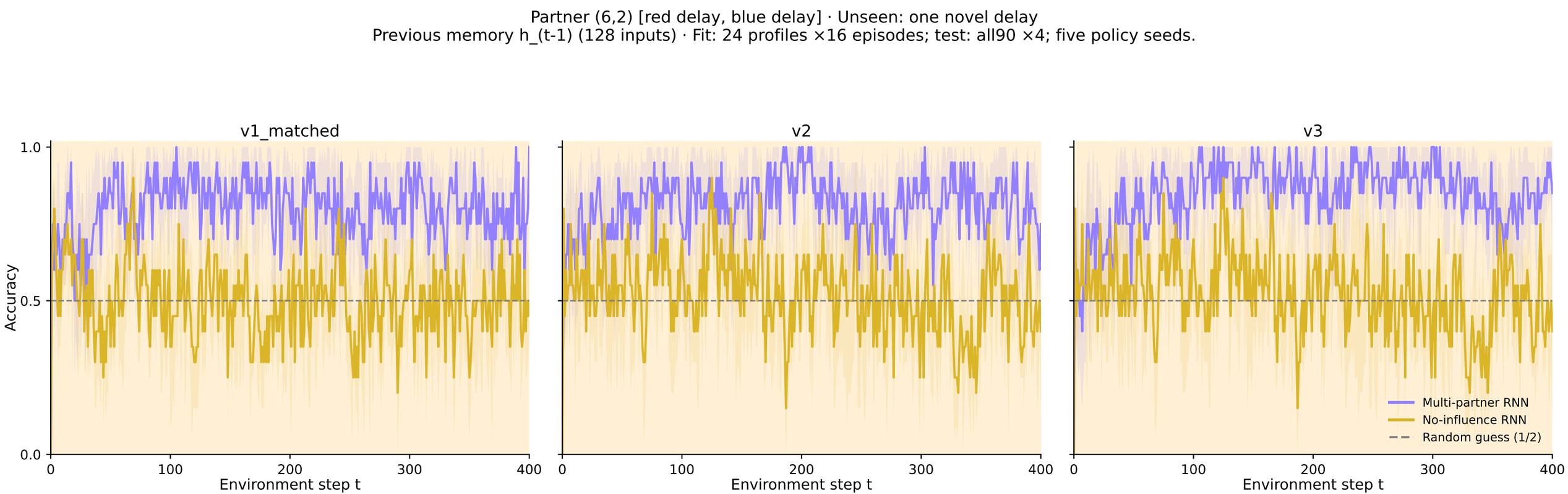

Partner (5,7) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

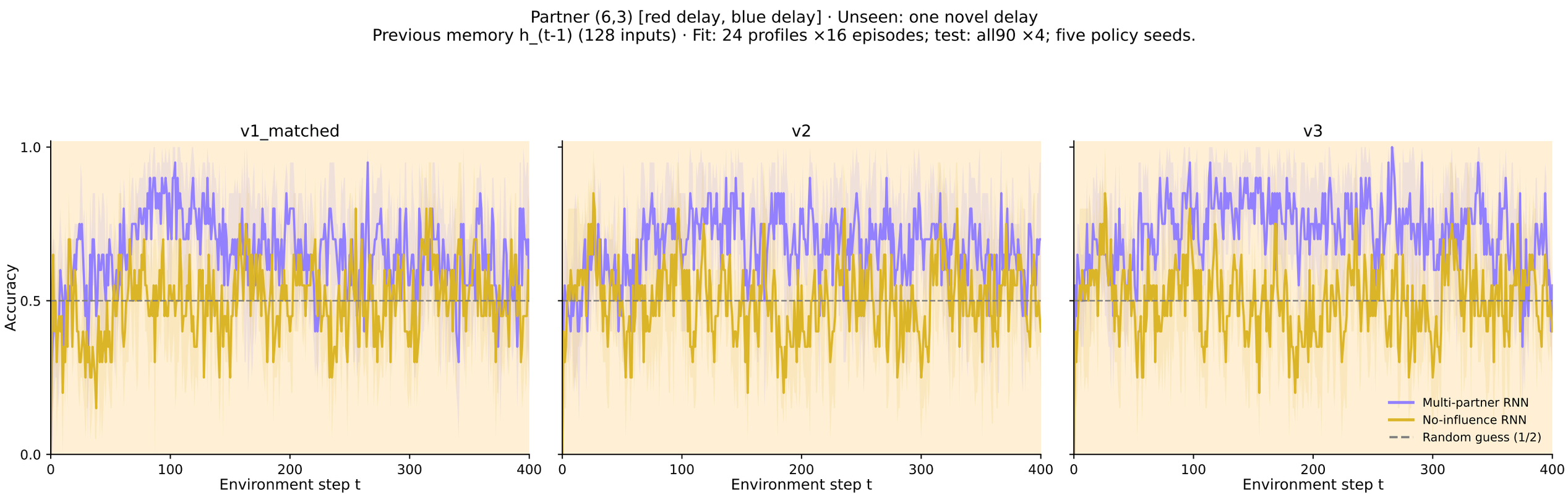

Partner (5,8) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

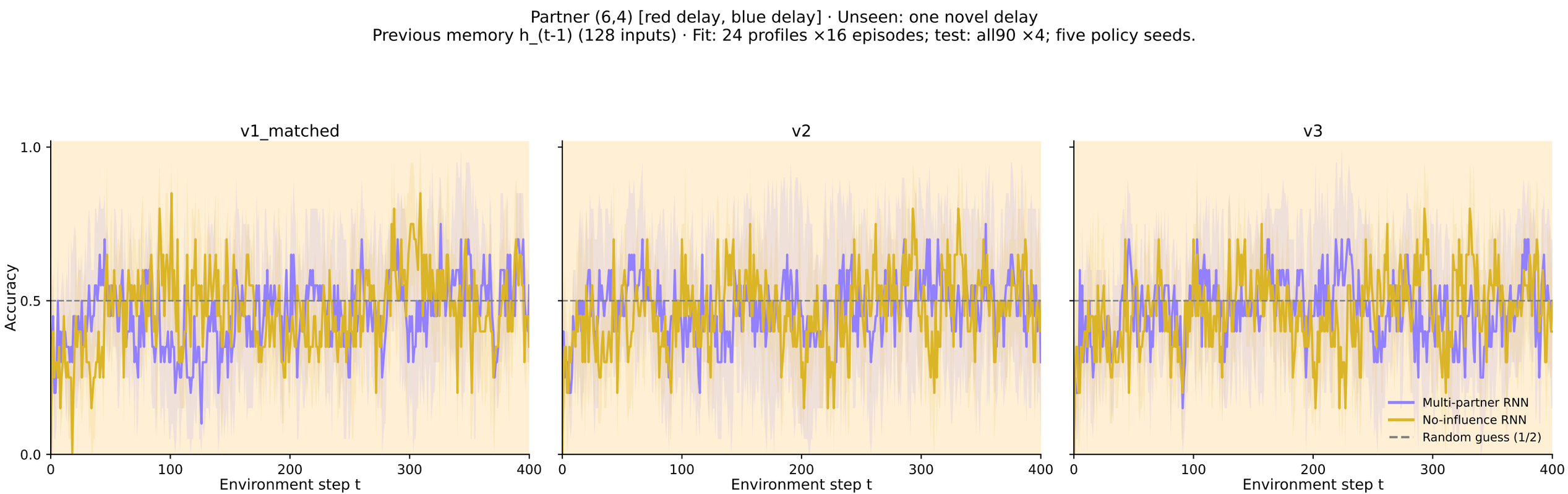

Partner (5,9) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

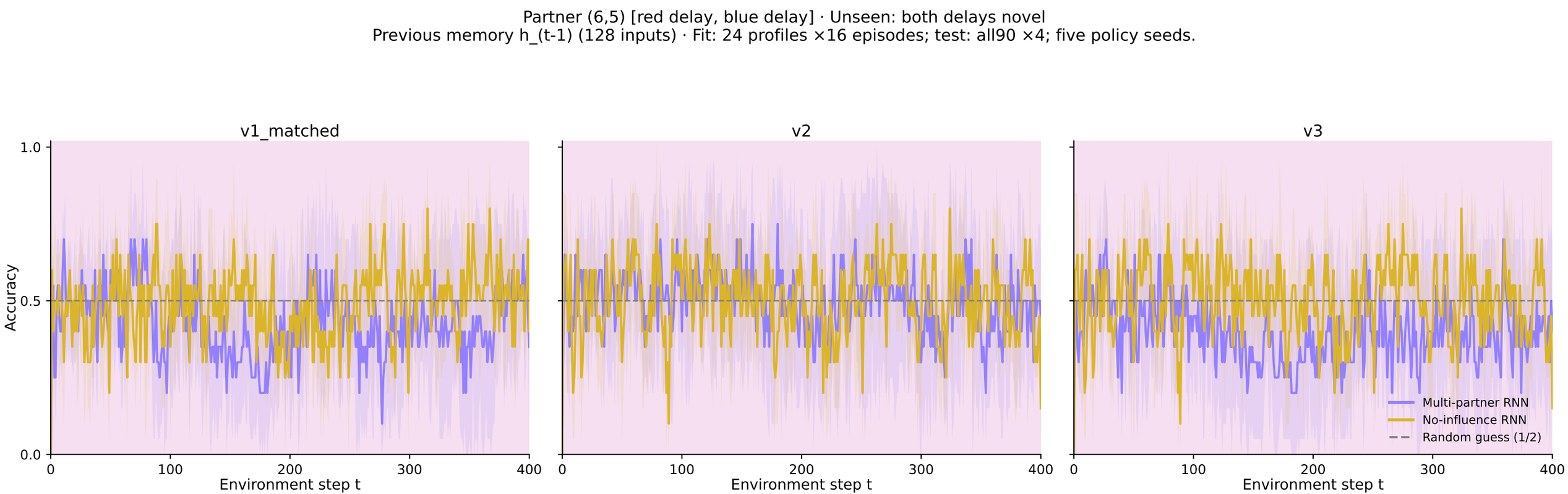

Partner (6,0) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

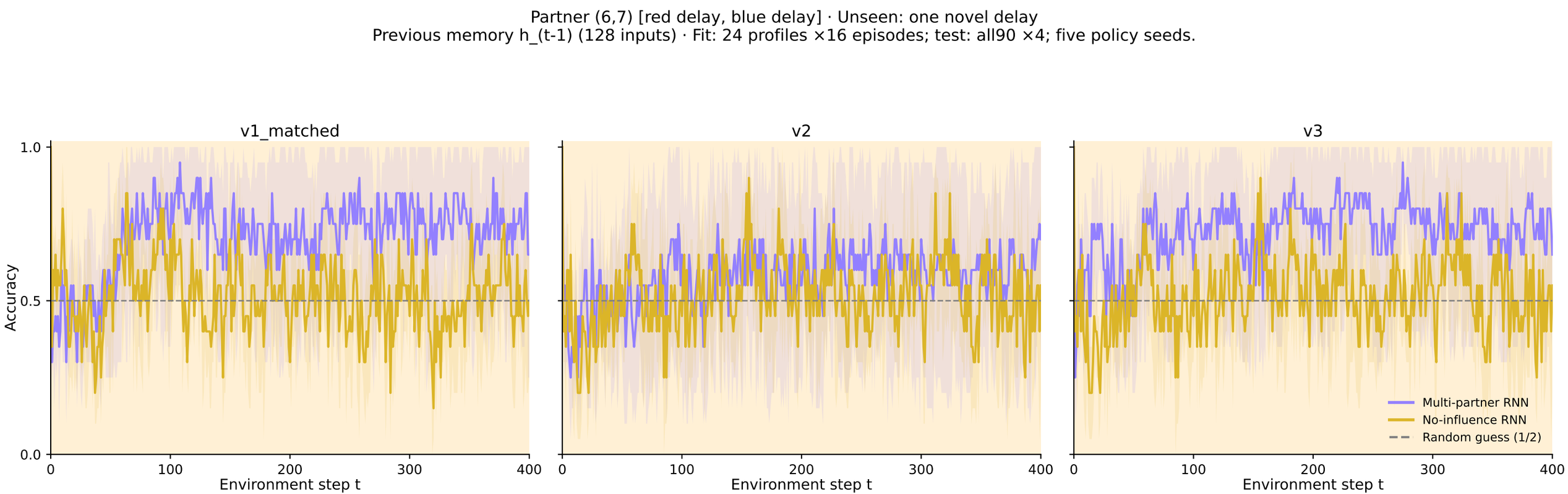

Partner (6,1) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

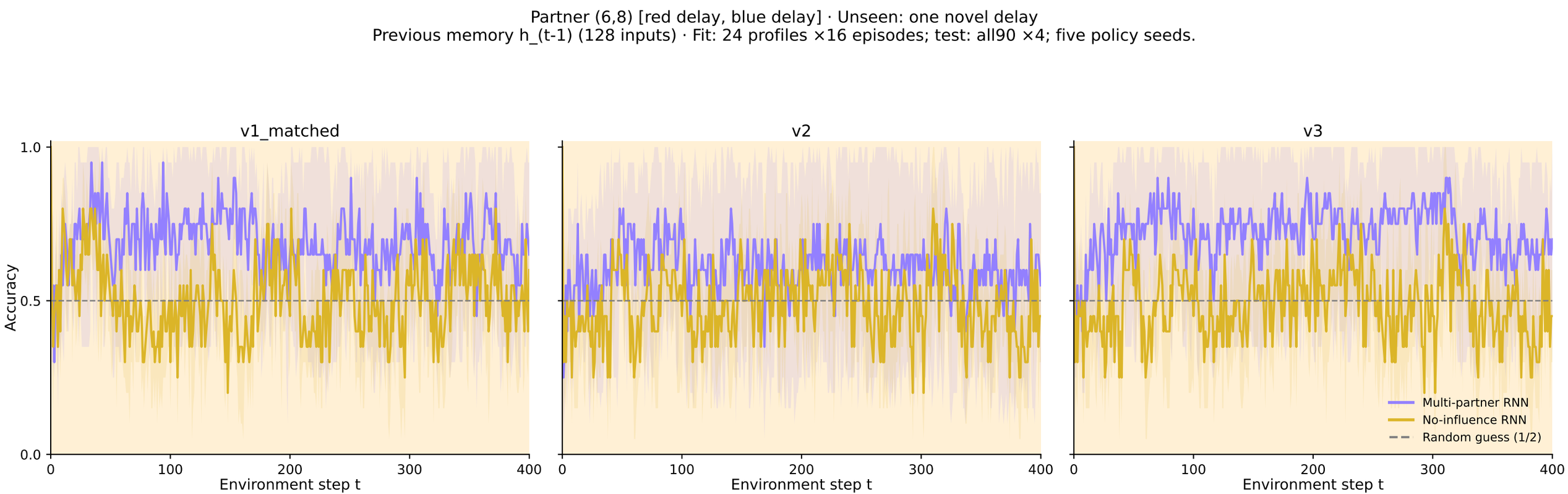

Partner (6,2) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

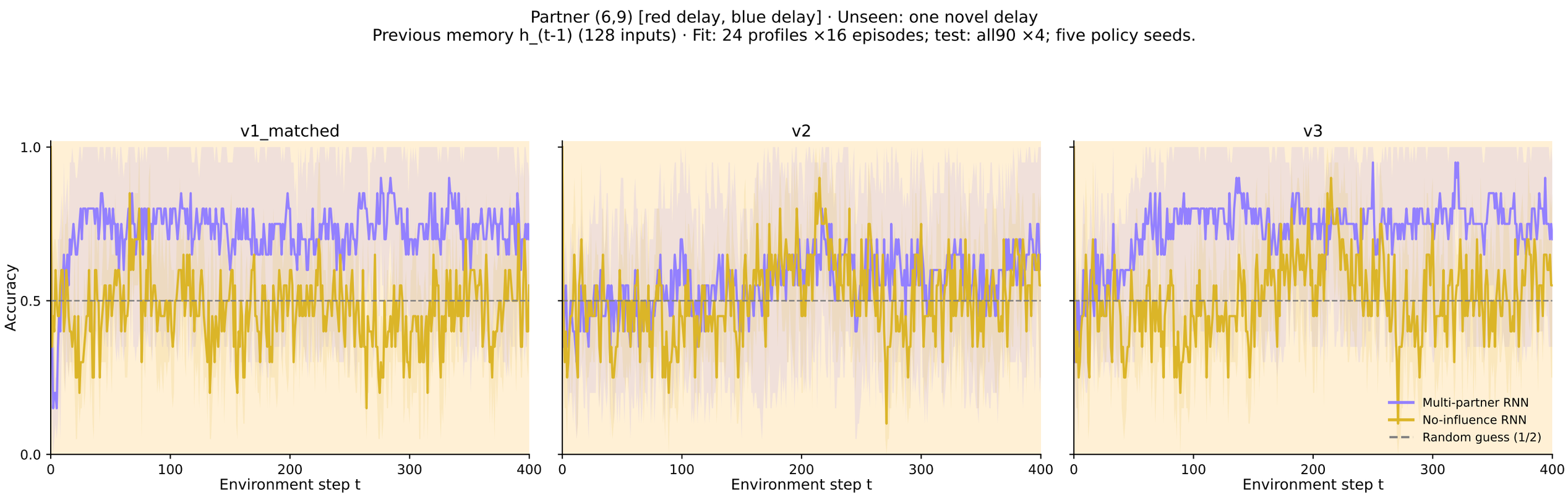

Partner (6,3) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

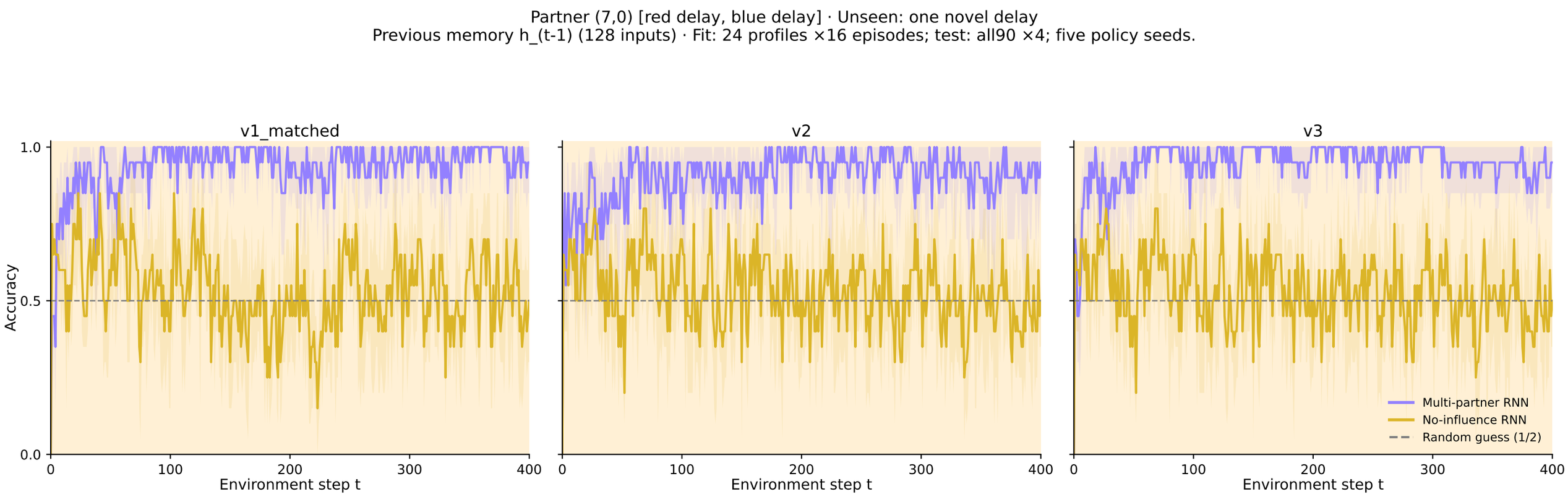

Partner (6,4) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

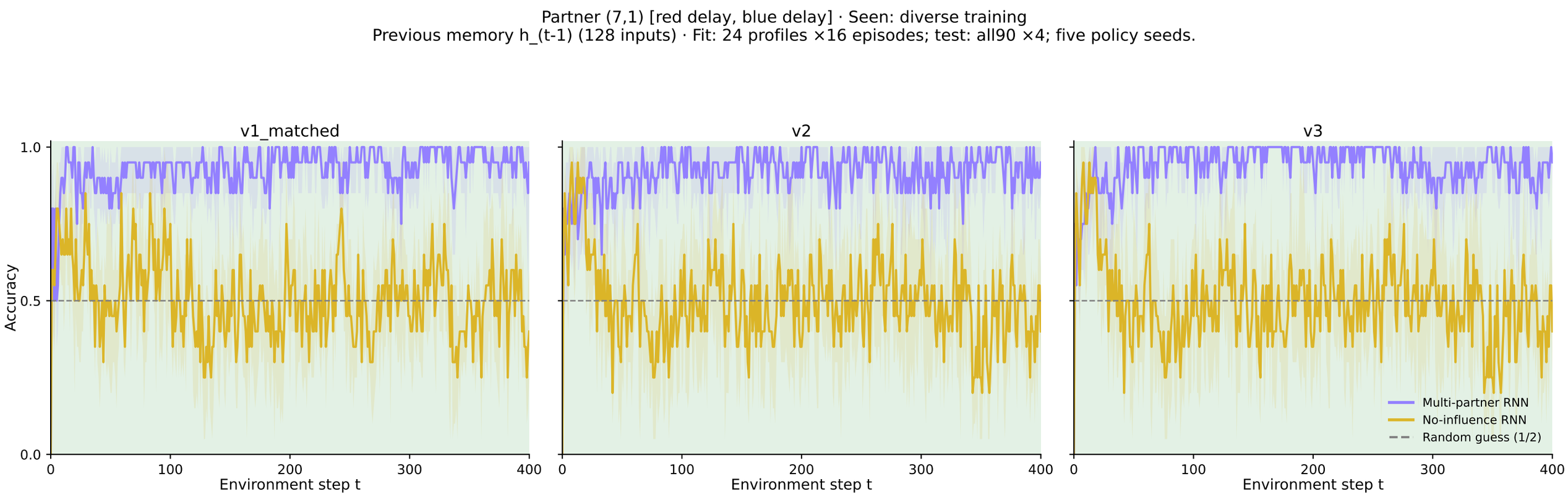

Partner (6,5) [red delay, blue delay] · Unseen: both delays novel
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

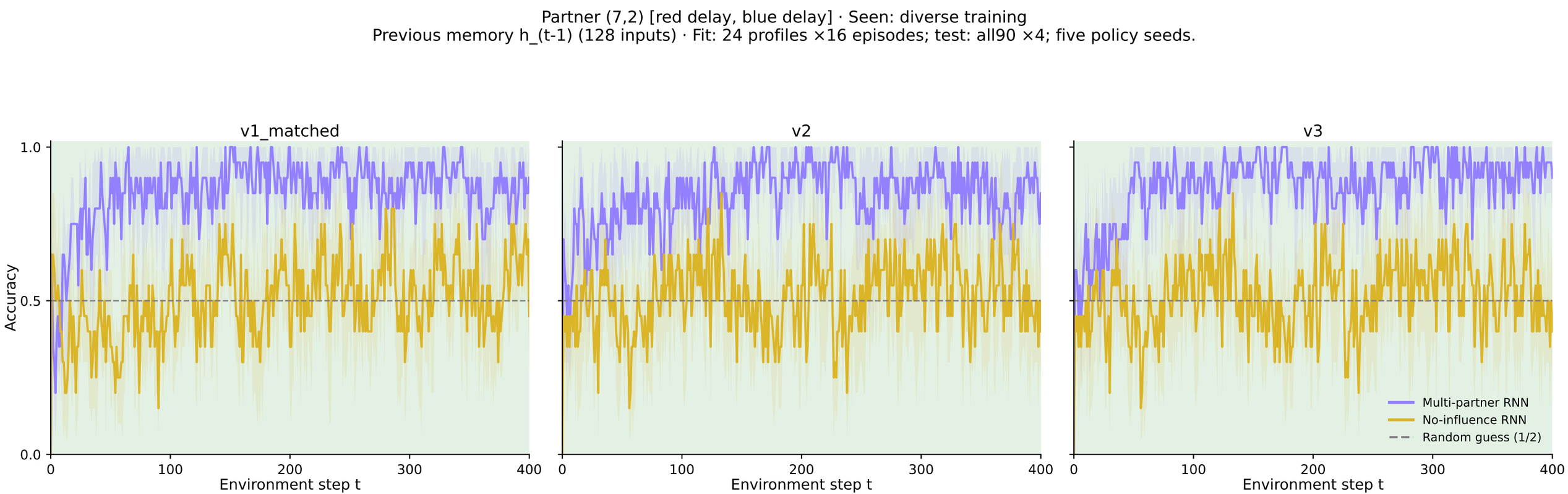

Partner (6,7) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

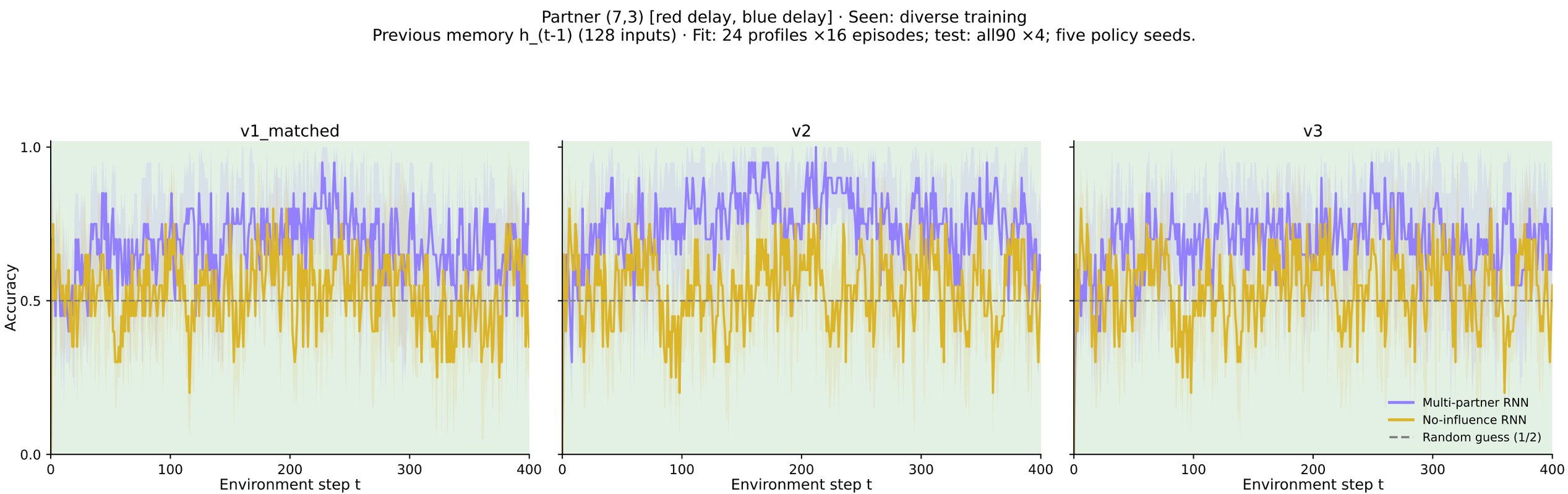

Partner (6,8) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

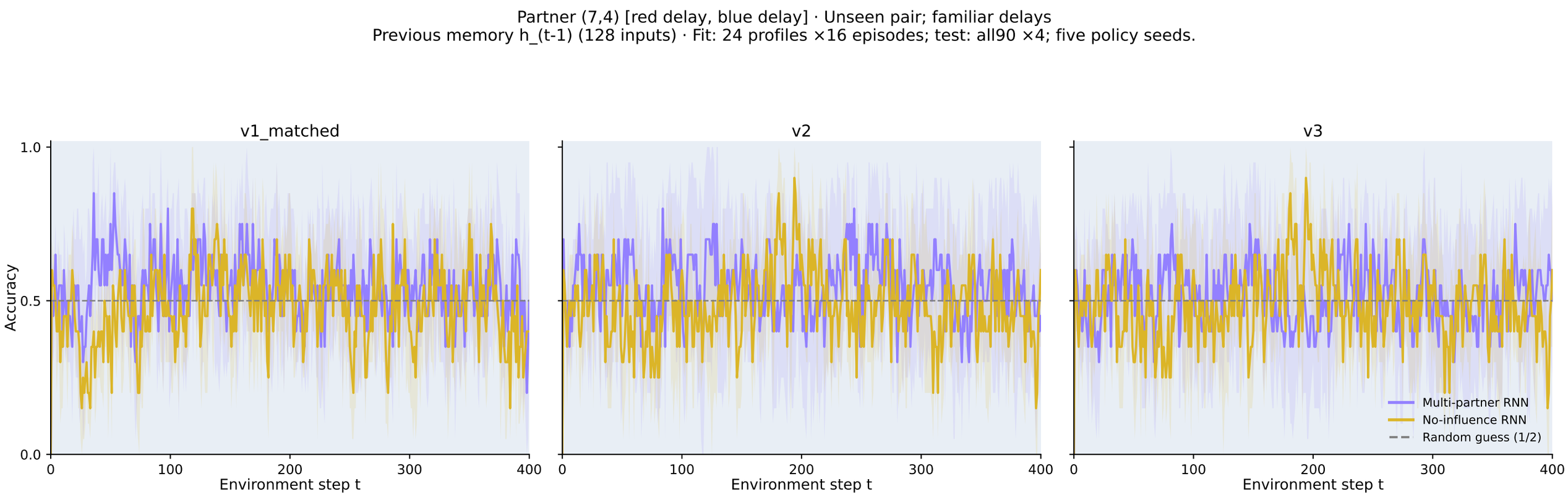

Partner (6,9) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

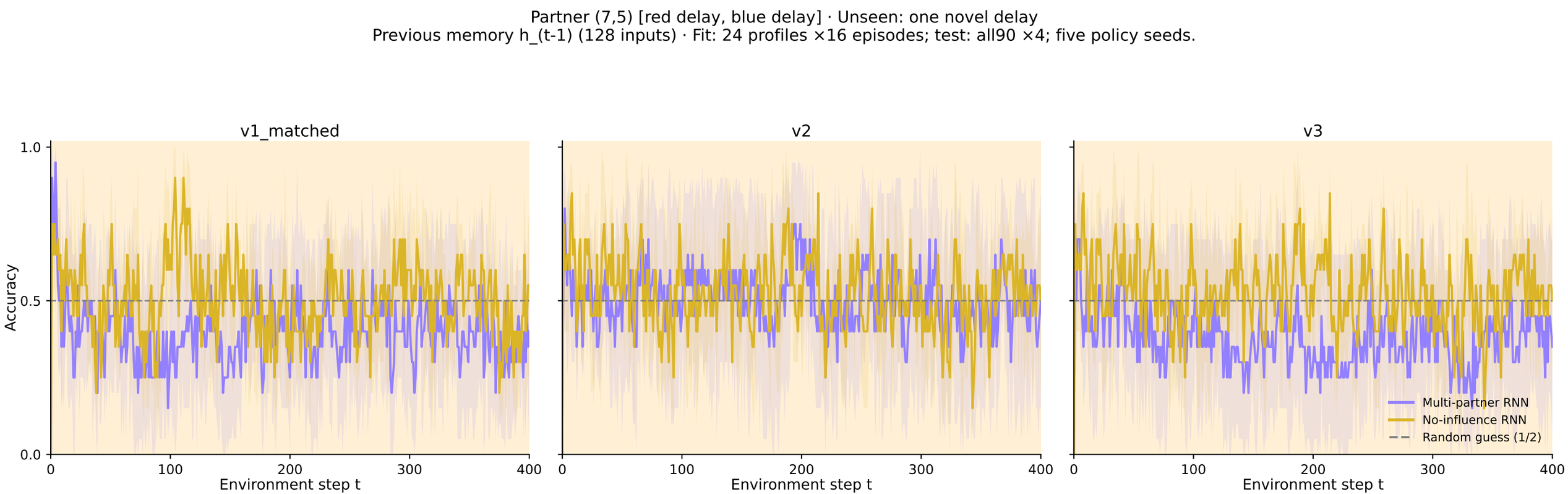

Partner (7,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

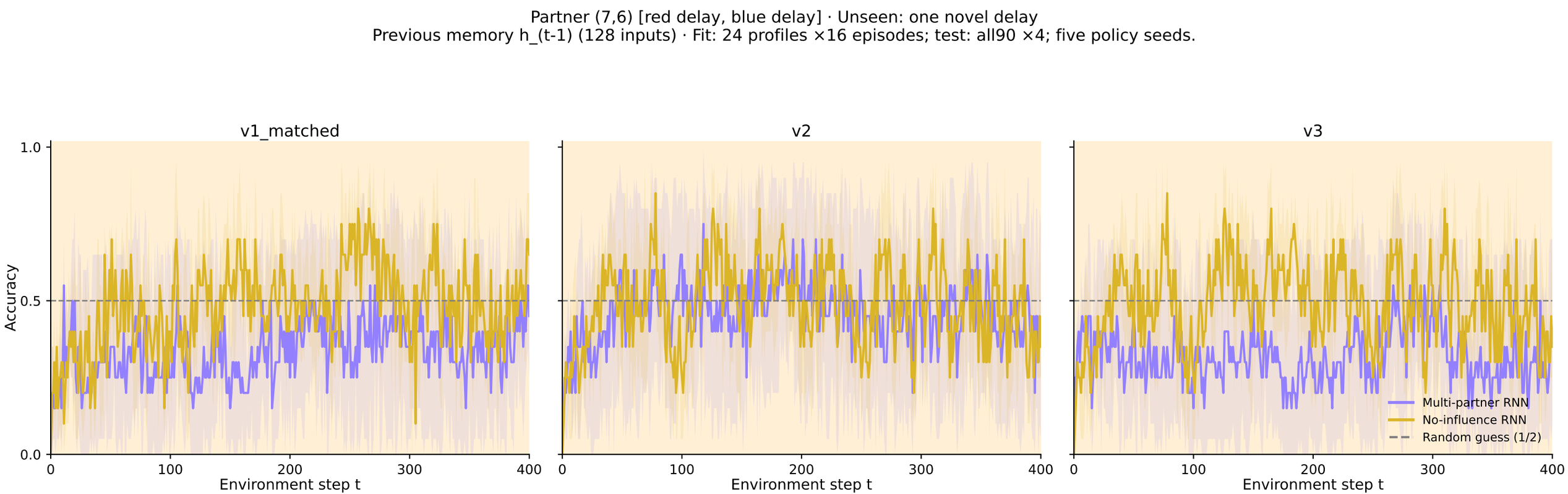

Partner (7,1) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

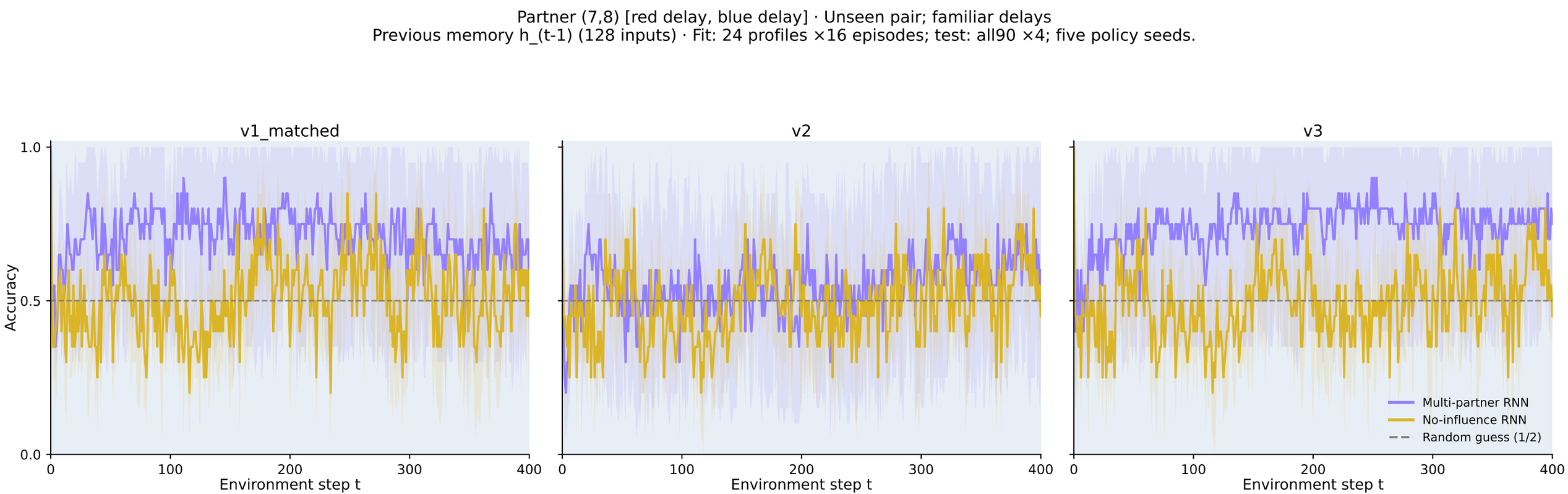

Partner (7,2) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

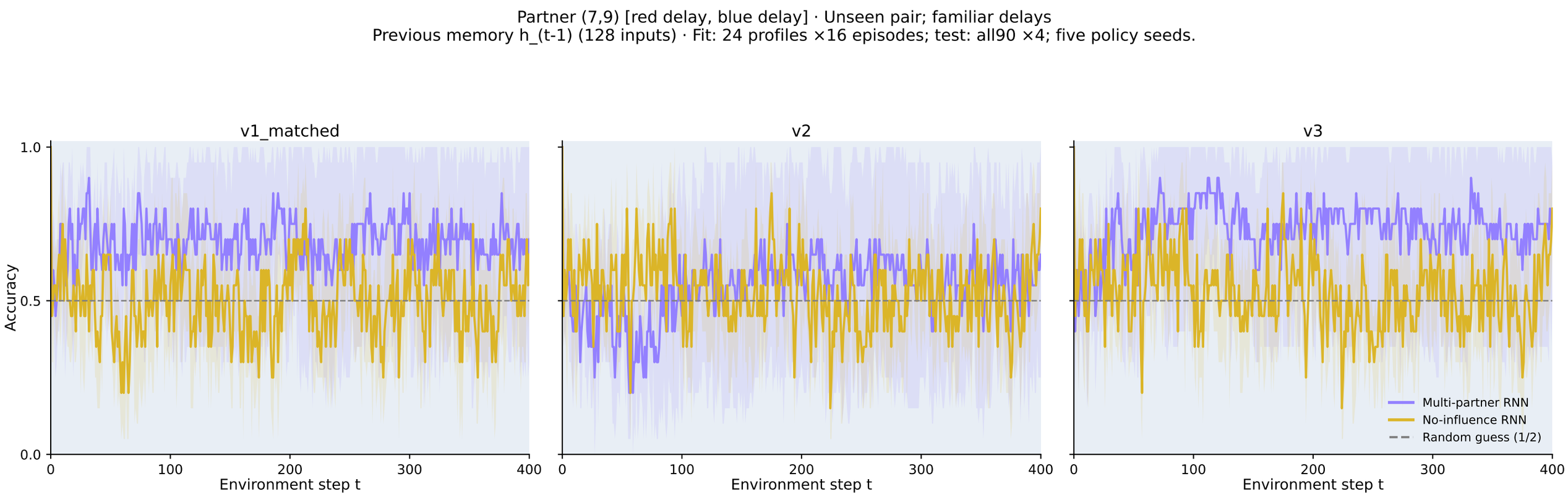

Partner (7,3) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

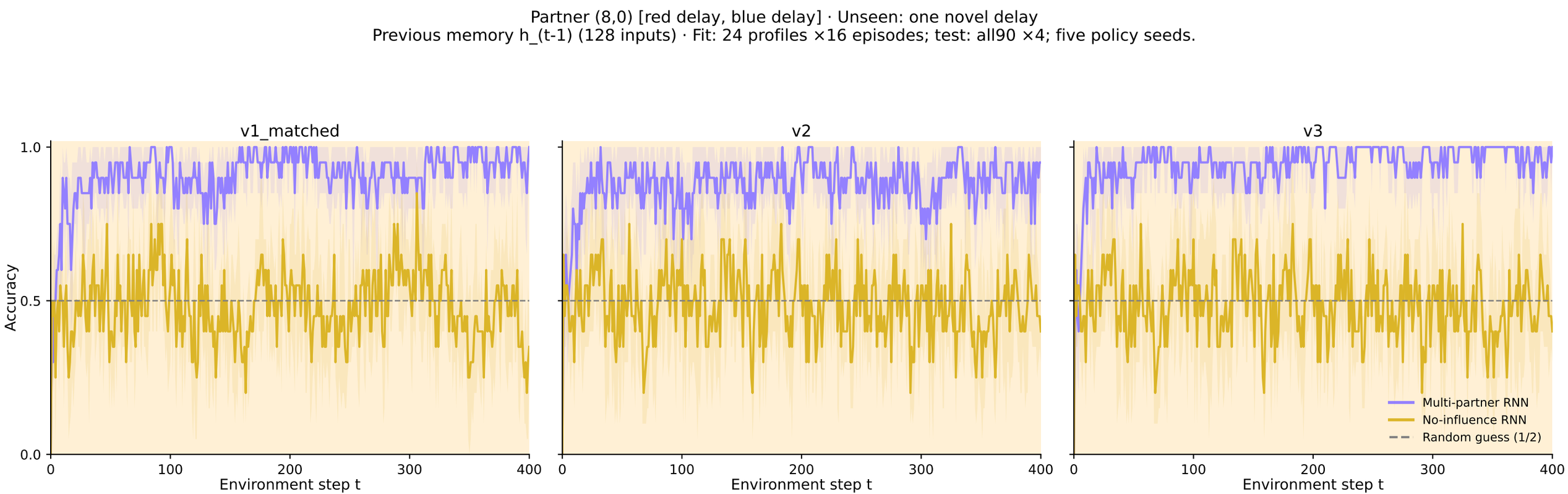

Partner (7,4) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

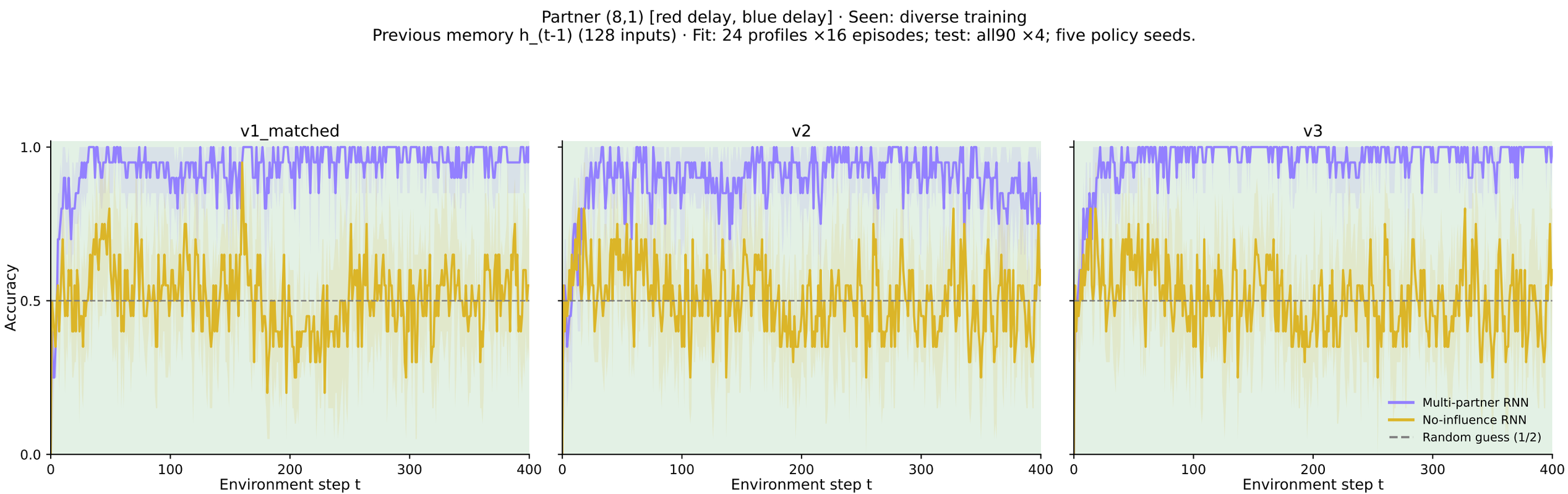

Partner (7,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

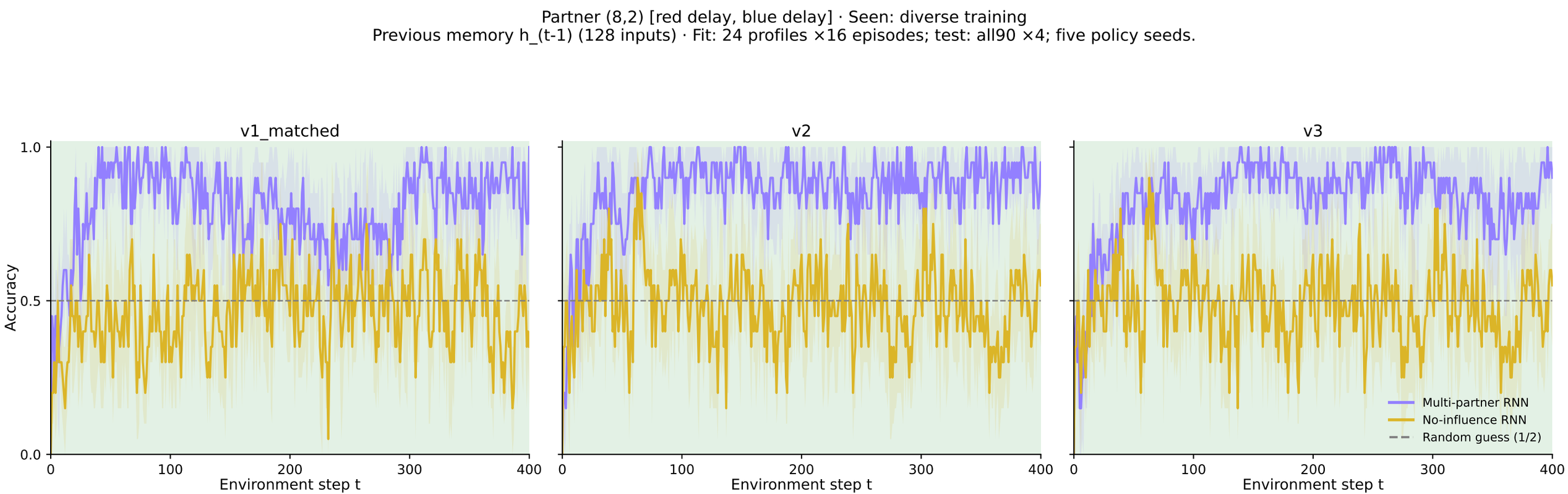

Partner (7,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

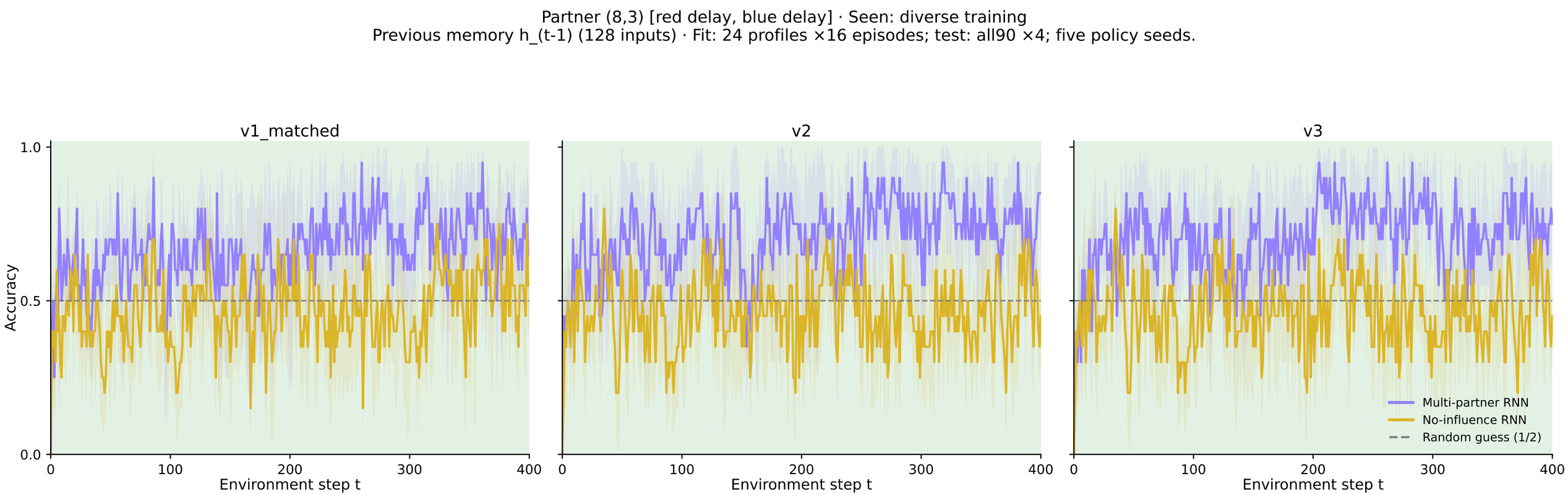

Partner (7,8) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

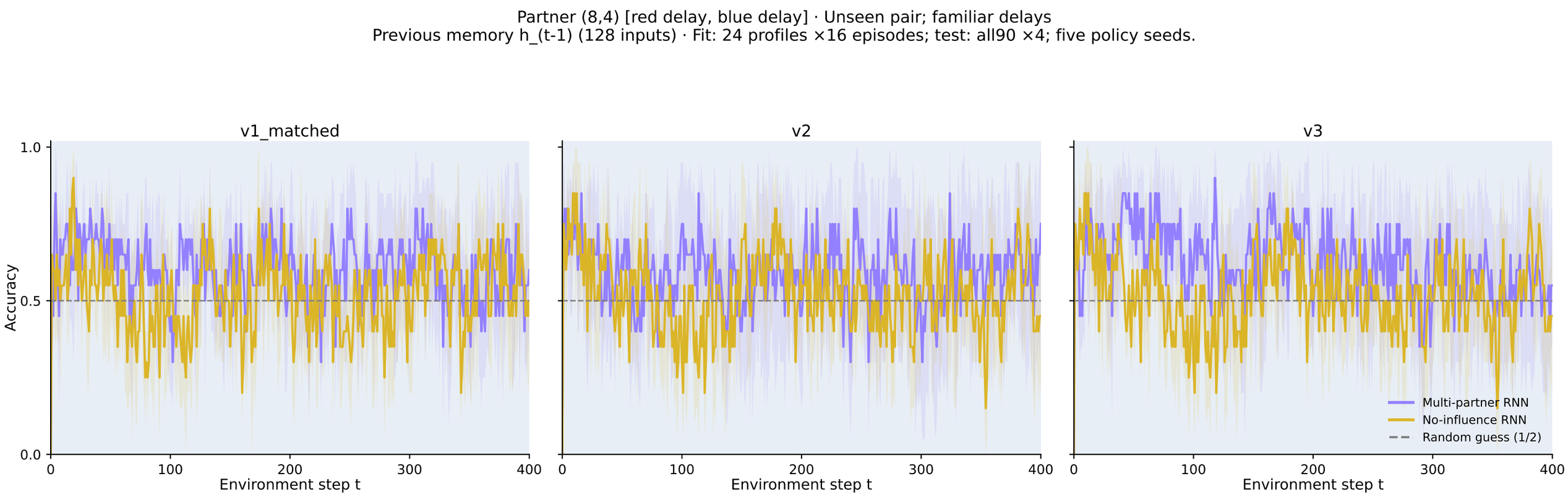

Partner (7,9) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

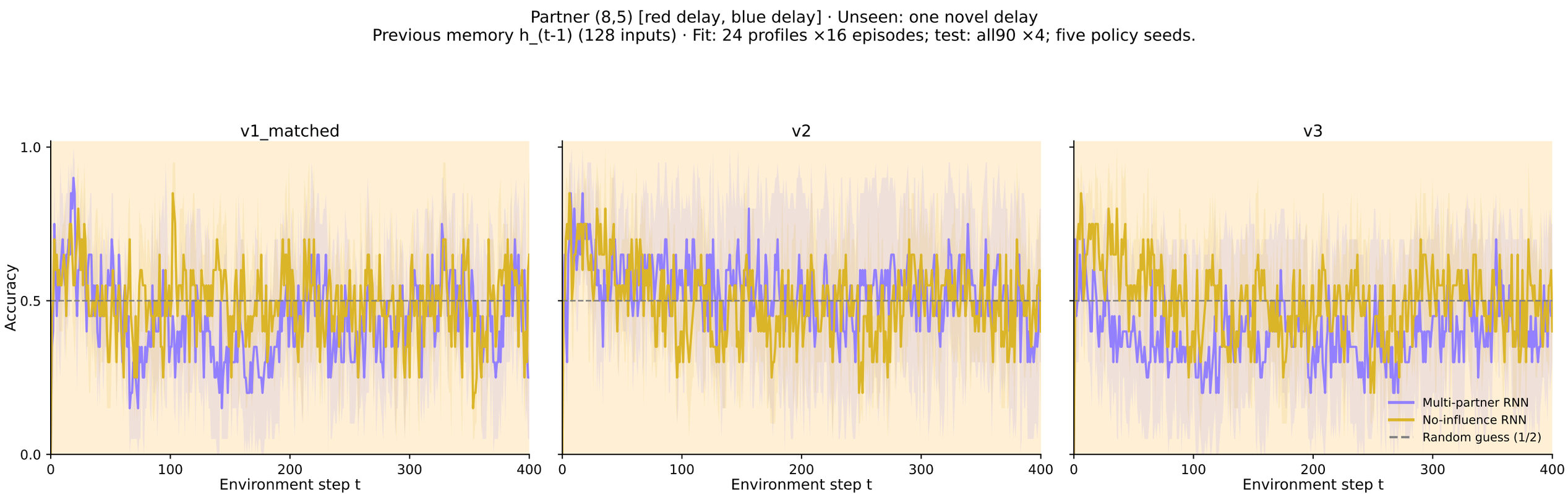

Partner (8,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

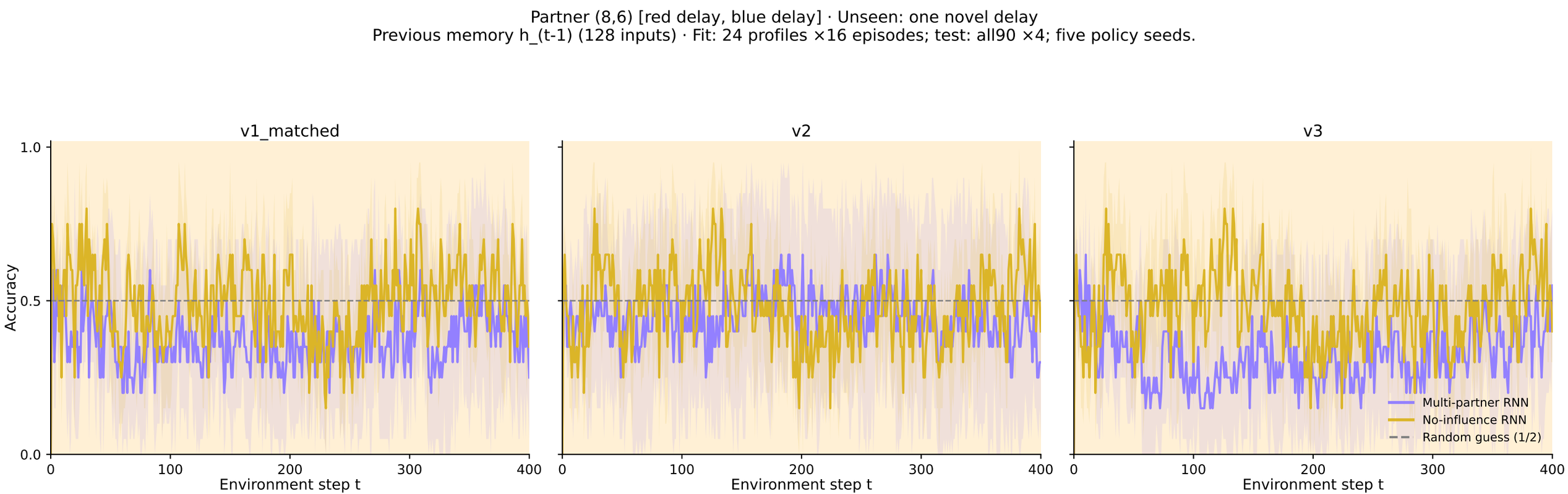

Partner (8,1) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

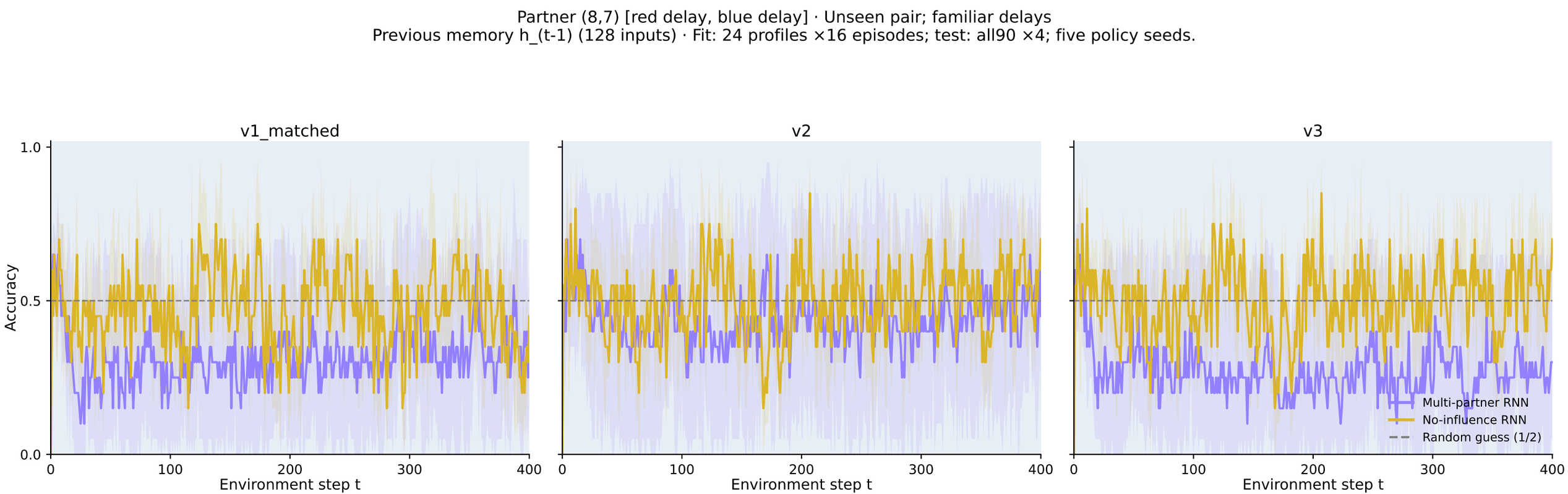

Partner (8,2) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

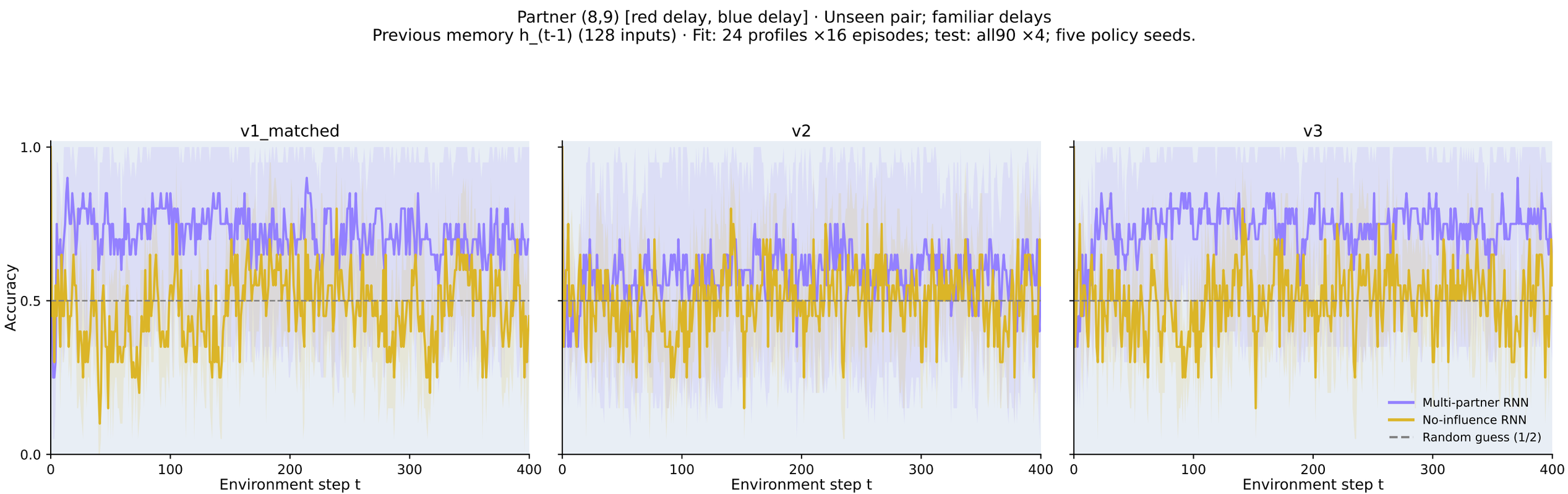

Partner (8,3) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

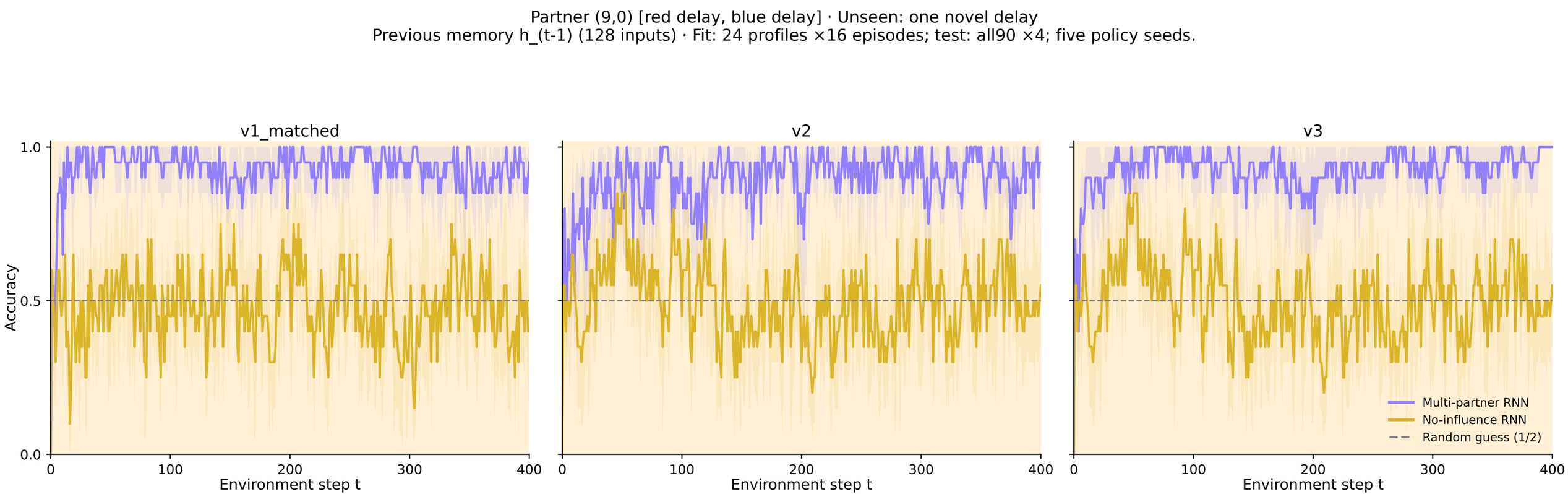

Partner (8,4) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

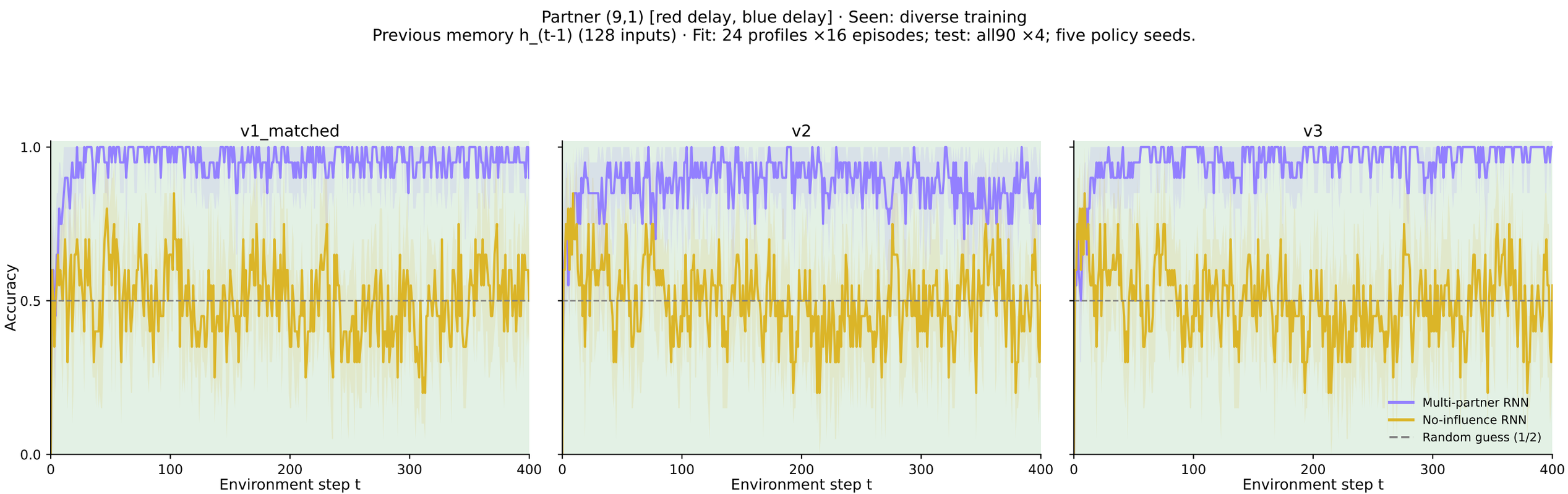

Partner (8,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

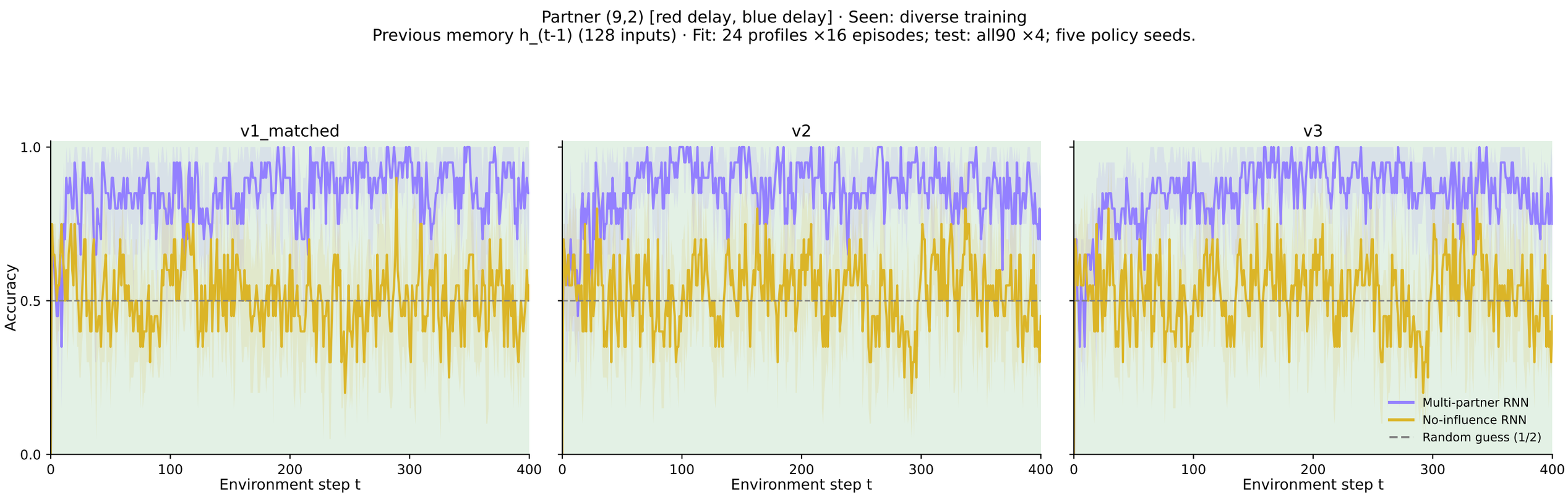

Partner (8,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

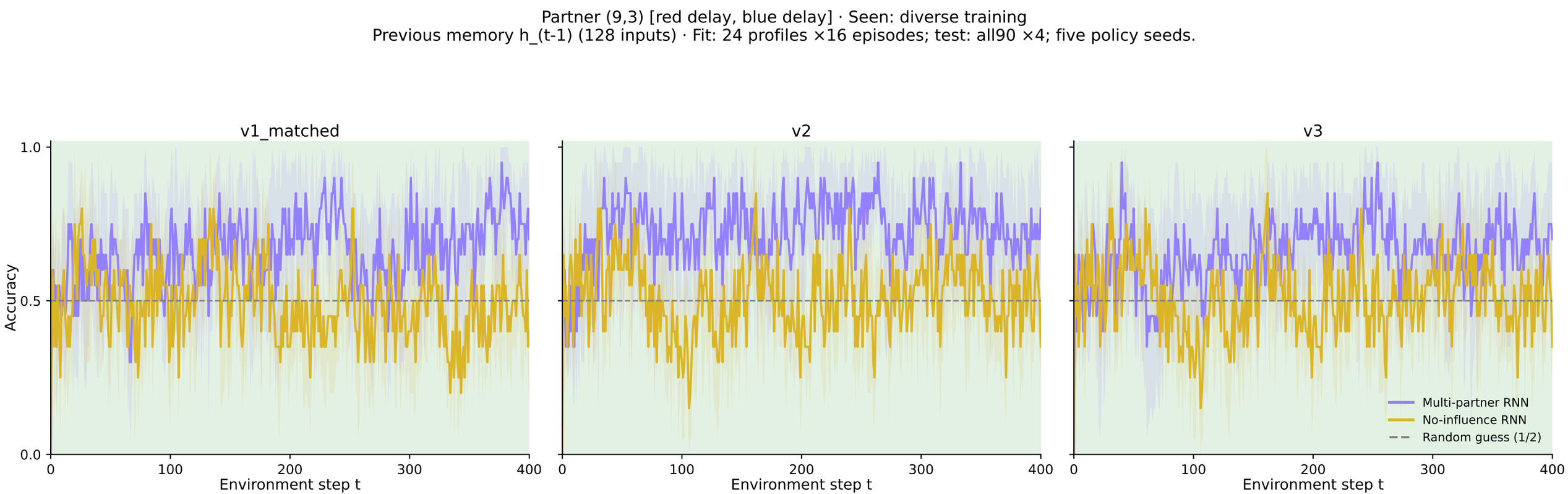

Partner (8,7) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

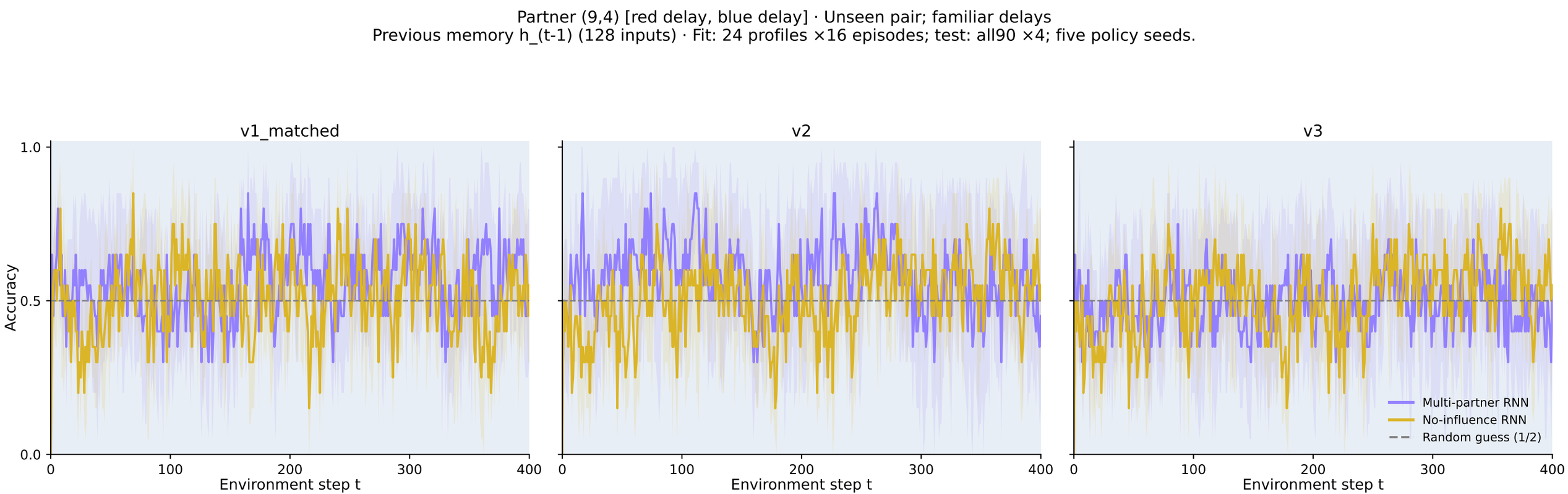

Partner (8,9) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

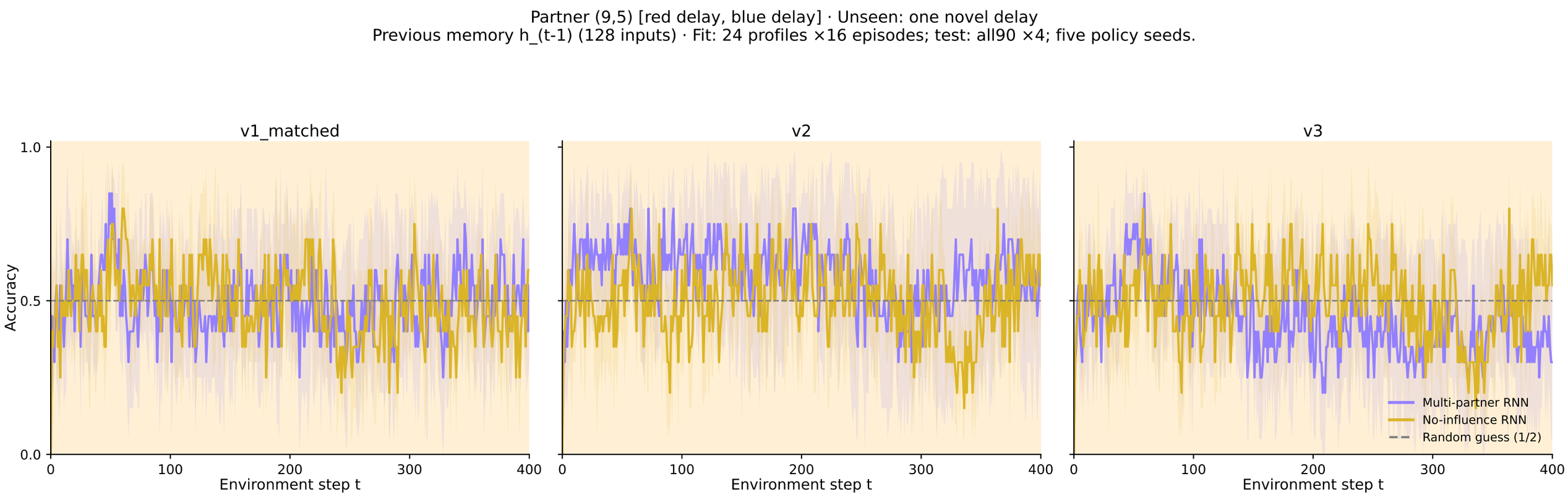

Partner (9,0) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

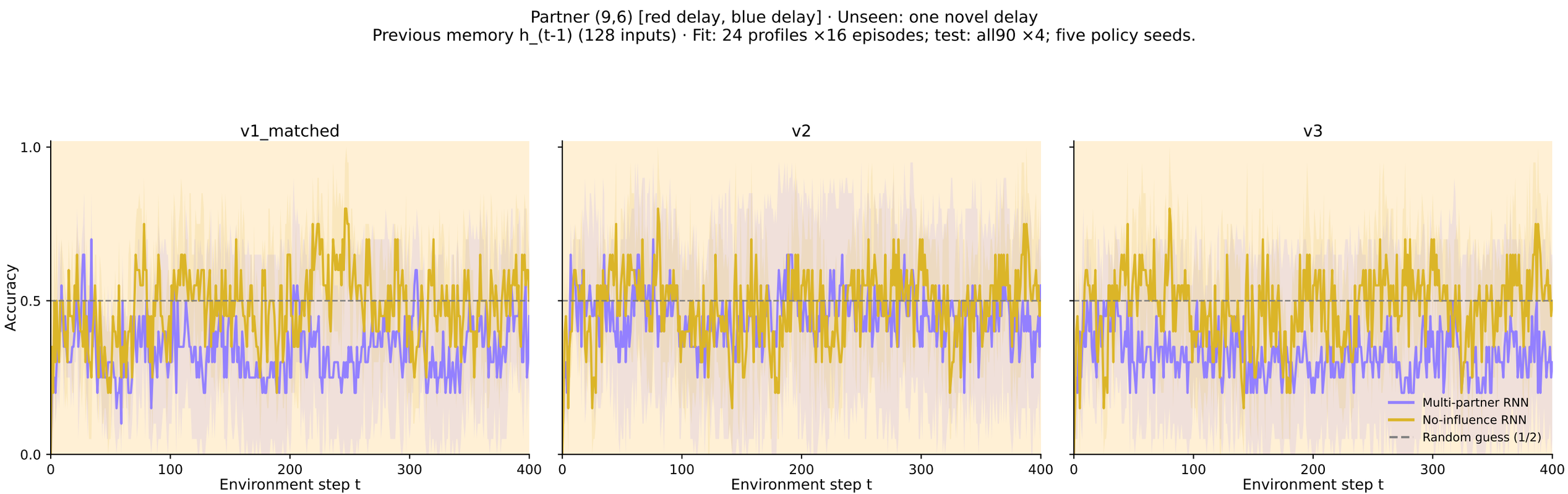

Partner (9,1) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

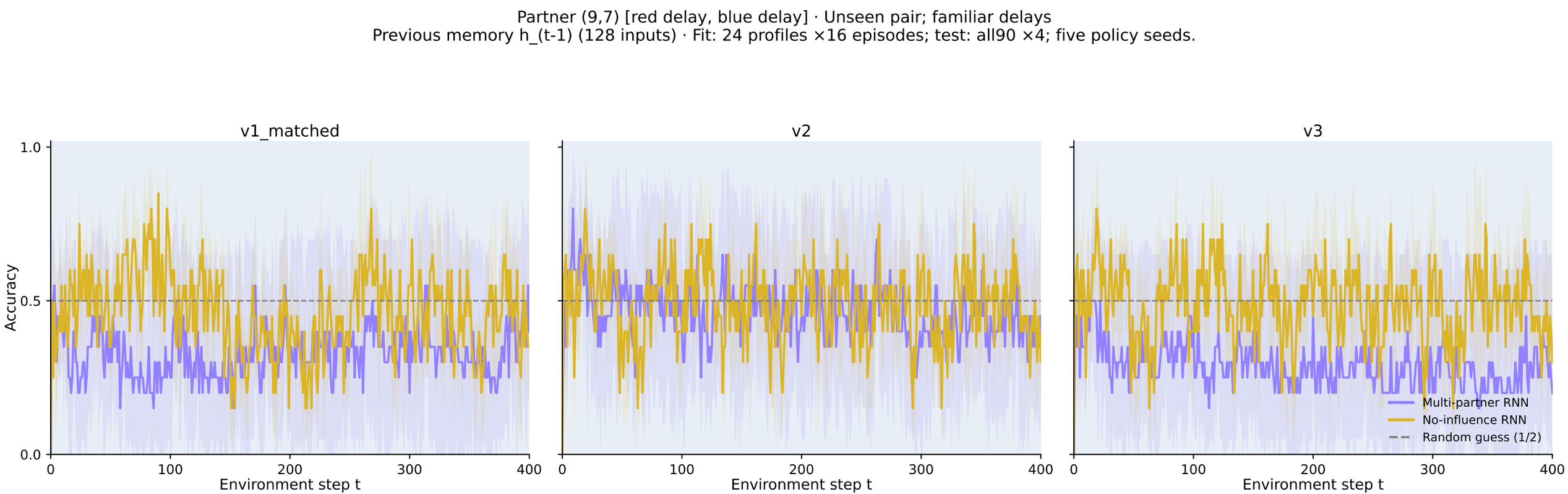

Partner (9,2) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)

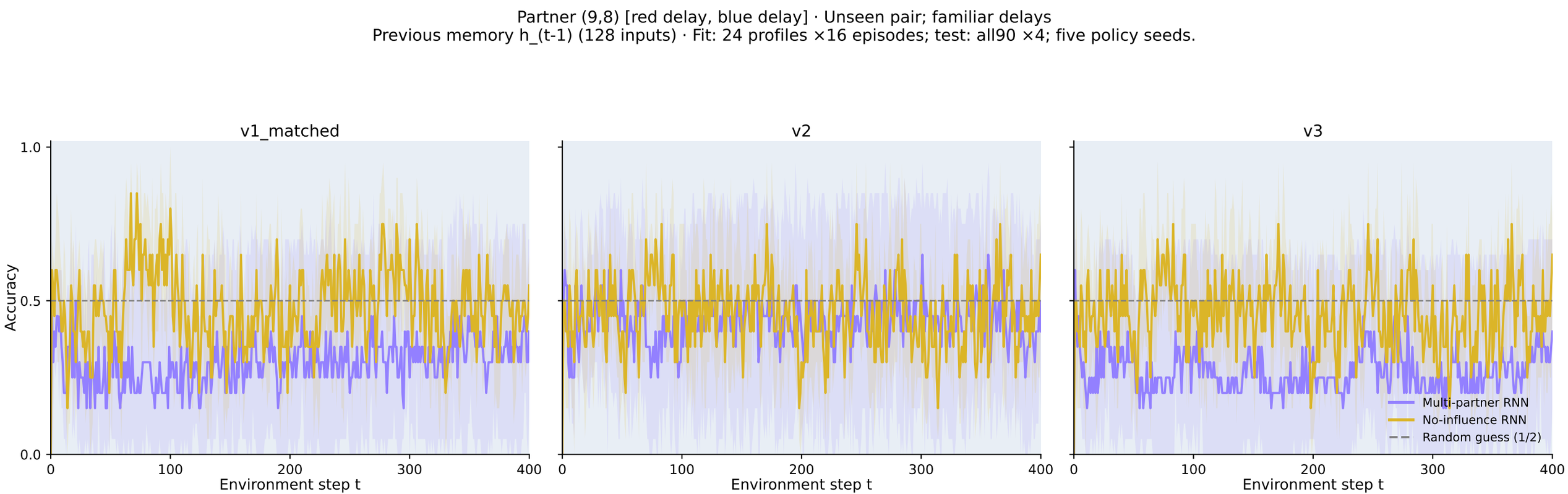

Partner (9,3) [red delay, blue delay] · Seen: diverse training
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (9,4) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (9,5) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (9,6) [red delay, blue delay] · Unseen: one novel delay
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (9,7) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)


Partner (9,8) [red delay, blue delay] · Unseen pair; familiar delays
Previous memory h_(t-1) (128 inputs) · Fit: 24 profiles ×16 episodes; test: all90 ×4; five policy seeds.
v1_matched
v2
v3
Accuracy
1.0
0.5
0.0
0
100
200
300
400
Environment step t
Multi-partner RNN
No-influence RNN
Random guess (1/2)